The Crinkle-Cut Supercube

Constraint-Directed Infolding, Hidden Surface Area,
and the Geometry of Emergent Boundaries

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Abstract

Ordinary origami specifies a transformation and treats the resulting boundary as its consequence: a sequence of folds, or the crease pattern that records their result, is given in advance, and the finished shape follows from executing it. This monograph describes and formalizes a different construction, in which a small number of relations the finished boundary must satisfy are specified instead, and the transformation that satisfies them is discovered rather than prescribed.

The central example is a physical construction, the Crinkle-Cut Supercube, in which three non-contiguous squares were marked on a single large sheet, each annotated with the edge and vertex roles it was to occupy in a finished cube corner. No crease pattern connected the three squares in advance. The sheet was folded, pleated, and where necessary crumpled until the three marked regions achieved their declared exterior relations, with the remainder of the sheet, considerably larger in area than the three faces it supports, folded into the object's interior rather than trimmed away.

The monograph argues that existing origami notation, built around sequences of operations or complete crease patterns, cannot describe a specification of this kind, and introduces a fourth descriptive level, the constraint level, together with a formal vocabulary of marked regions, local frames, boundary grammars, and a constrained search operator, sufficient to state such a specification precisely. This formalism is then used to establish the monograph's central theoretical claim: that a finished boundary satisfying a declared specification does not, and structurally cannot, disclose which of the many interior histories capable of satisfying that same specification actually produced it. Exterior equivalence, in the sense the monograph makes precise, does not imply structural, crease, or procedural equivalence among the realizations that share it.

The argument is extended to the embodied character of the search process itself, to a carefully bounded structural analogy with the folding of the cerebral cortex, and to a formally hedged principle describing the conditions under which spatially separated, locally framed regions can be brought into a shared exterior configuration without cutting the sheet that connects them. Throughout, the monograph distinguishes what a single constructive

demonstration actually establishes from what remains conjectural, and treats the finished, visibly irregular object not as an imperfect execution of a clean design but as a physical certificate that a sparse specification, realized by a much larger and largely unrecoverable hidden process, was in fact satisfiable.

Contents

Introduction: The Boundary Before the Fold

Ordinary origami begins with a transformation and ends with a shape. A sequence of folds, or a crease pattern that records their result, is specified in advance, and the folder's task is to execute it faithfully. The final boundary of the object is a consequence of the folds, not their premise.

Constraint-directed infolding inverts this order. Instead of specifying the transformation and observing what boundary results, the constructor specifies a small number of relations the final boundary must satisfy and permits the transformation itself to be found rather than prescribed. Selected regions of a sheet are marked and assigned roles: this edge is to become a bottom edge, that one a side edge, these three corners are to meet at a shared vertex. Nothing is specified about how the material between those regions is to be folded, or in what order, or with what crease pattern. It is left free to buckle, pleat, invert, overlap, or crumple, provided that whatever it does, the marked relations end up satisfied.

The Crinkle-Cut Supercube is the constructive example around which this monograph is built. Three non-contiguous squares were marked on a single large sheet, at some distance from one another and with no shared edge. Each square carried edge annotations indicating which of its sides should occupy the bottom, and which the side, of a finished cube. The sheet was then folded, without a predetermined crease pattern, until those three marked squares had been brought together as three mutually adjacent faces sharing a common corner. The remainder of the sheet, considerably larger in area than the three faces it was made to support, was not cut away. It was pushed, pleated, and crumpled into the interior of the resulting box, where it remains, hidden but present, as the material cost of the exterior that is visible.

This is not a polished folding technique, and it is not offered as one. The finished object is visibly irregular. Its value lies elsewhere: it demonstrates that a target exterior can be specified more sparsely than a complete crease pattern requires, and that the gap between a sparse specification and a complete physical object can be closed by permitting unconstrained material to absorb whatever geometric surplus the specification does not resolve. What is prescribed is preserved. What is not prescribed is not thereby prevented; it is simply free, and the sheet finds *some* way to accommodate it.

The central claim developed across the monograph is the following:

> *Constraint-directed infolding preserves selected surface distinctions by transferring unresolved geometric surplus into hidden interior structure.*

Everything that follows is an attempt to make this claim precise, to give it a formal vocabulary, to trace its consequences for what a finished boundary can and cannot disclose about its own history, and to ask how far a strategy discovered by hand with a single sheet of paper generalizes to other systems in which a legible exterior is achieved atop a much larger, much less legible interior.

0.1 Scope and disclaimers

Three things this monograph does not attempt should be stated plainly before the argument begins.

It does not offer a universal folding theorem. The claims advanced here are existence claims for a demonstrated class of constructions, not general theorems about which boundary specifications are realizable from which sheets. Where a stronger conjecture is stated, it is marked as a conjecture and not treated as established.

It does not offer a complete physical theory of crumpling. Crumpling is treated throughout as a name for a family of admissible deformations whose exact microstructure is left underdetermined by design, not as a phenomenon whose mechanics this monograph resolves. Existing work in the physics of crumpled sheets is drawn on where relevant but is not the subject of the monograph.

It does not offer a literal model of cortical development. A structural analogy is developed in Part V between the paper construction and the folding of the cerebral cortex, because both involve an area-rich surface accommodating itself to a volume-limited environment. That analogy is developed carefully and its limits are stated explicitly, in the same chapter in which it is introduced, precisely because the temptation to overstate it is real and the biological literature does not support a literal reading.

0.2 How the argument proceeds

The monograph has three movements.

The first movement, comprising Part I, is constructive. It reconstructs the original experiment in enough detail to establish exactly what was demonstrated, and it identifies precisely why existing origami notation, built around operations and crease patterns, cannot describe what the experiment did. This motivates the introduction of a fourth descriptive level, above operations, creases, and shapes: the level of constraints.

The second movement, comprising Parts II and III, is formal. A vocabulary of marked regions, local frames, boundary conditions, and a search-like infolding operator is developed in enough detail to state the Supercube's specification precisely. This formal apparatus is then used to show why a satisfied specification cannot, by inspection of the exterior alone, disclose the interior history that satisfied it. This is the monograph's central theoretical result, and it follows directly from the formalism rather than being asserted independently of it.

The third movement, comprising Parts IV through VI, extends the argument outward: to the embodied, externally-scaffolded character of the search process itself; to a carefully bounded structural analogy with cortical folding; and finally to a general principle, stated with appropriate hedges, about when spatially separated marked regions can be brought into a shared exterior configuration without cutting the sheet that connects them.

The finished Supercube is, in the end, treated as a witness rather than as a specimen. It does not prove that every boundary specification is realizable. It proves that at least one nontrivial specification was, and it does so in a form that can be inspected, unfolded, and reasoned about directly. The rest of the monograph is an attempt to say, as precisely as possible, what that single physical fact actually shows.

Part I

The Constructive Problem

Chapter 1

A Cube Without a Net

1.1 The ordinary cube net as a solved problem

Before describing what the Crinkle-Cut Supercube did, it is worth being precise about what an ordinary folded cube does not have to do.

A standard cube net is six squares arranged in the plane so that each pair of squares meant to become adjacent faces already shares an edge in the unfolded layout. The classic cross-shaped net is one such arrangement among eleven possible ones, but every valid net shares the same property: the planar adjacency of its squares has already been chosen to match the adjacency the finished cube requires. Folding along the marked edges does not discover this adjacency. It executes it. The hard part of the problem, choosing which squares should touch which, has already been solved on paper before a single fold is made.

This is worth stating because it is easy to mistake the appearance of a folding task for its actual content. A net looks like raw material waiting to be shaped, but it is closer to a solution waiting to be unpacked. The folder's contribution is real but narrow: fold accurately along the given lines, in a sensible order, and the geometry follows automatically because it was arranged to follow automatically.

1.2 A different starting point

The Crinkle-Cut Supercube began from a different kind of specification, one that did not solve the adjacency problem in advance.

A single large sheet, substantially bigger than the six-square area a minimal net would require, was the starting material. Three squares were marked on this sheet, and critically, the three squares were not adjacent to one another and did not share edges. They sat at separated locations across the sheet, connected only through the intervening material that lay between them.

Each square was annotated along two of its edges to indicate the role that edge was meant to play in the finished object. One edge was marked to become a bottom edge of the eventual cube; another was marked to become a side edge. A third mark, shared conceptually across the three squares even though it appeared as a separate small annotation on each, indicated that a particular corner of each square was meant to become the same corner of the finished object; the three squares were to meet, eventually, at one shared exterior vertex.

Nothing else about the sheet was specified. No crease pattern connected the three squares to one another. No instructions indicated how the intervening material should be treated, in what order the folds should proceed, or what shape that material would end up taking. Only the exterior relations among the three marked squares were declared: which edges belonged where, and which corners were to coincide.

1.3 What the marks meant

It is worth dwelling on what kind of information the blue markings actually carried, because it is easy to mistake them for crease instructions and they were not that.

A crease instruction says: fold here, in this direction. The blue marks on the Supercube's three squares said something different: this edge, wherever it ends up and however it gets there, must be the bottom edge; this other edge must be the side edge; this corner must coincide with these two other corners. The marks were coordinate annotations, not motion instructions. They fixed a target relation among features of the sheet without fixing the path by which that relation would be reached.

This distinction turns out to matter a great deal, and later chapters will develop it formally. For now it is enough to notice that the three marked squares functioned like local frames: each one carried its own small coordinate system, indicating which of its edges corresponded to which role in the eventual object, independent of where that square happened to sit on the original sheet or how far it was from the other two.

1.4 The construction

The construction proceeded, as closely as the surviving photographs and the constructor's account allow it to be reconstructed, through four rough stages.

Marking. The three squares were identified on the flat sheet and their edges annotated as described above. At this stage the sheet was still entirely flat, and the three squares were visibly separated by substantial intervening area, some of it many times the area of the squares themselves.

Approximate alignment. The sheet was folded, without a predetermined sequence, in whatever way brought the three marked squares closer to one another and closer to the relative orientation their annotations demanded. This stage involved a great deal of trial: a fold that brought two squares closer together might leave them misoriented relative to each other, requiring a different fold, or an additional one, to correct the orientation without losing the proximity already gained.

Corner formation. At some point in this process, the three squares arrived at a configuration in which their marked corners could be brought together at a single shared vertex, with their bottom and side edges correctly oriented relative to one another. This was the decisive moment in the construction. Once achieved, it gave the object a partial cube frame: three faces, correctly adjacent, correctly oriented, sharing a corner, even though the object as a whole was still far from resembling a finished cube.

Infolding of the remainder. The material that had not been selected as one of the three faces, still connected to all three of them and still comprising the great majority of the sheet's total area, was folded, pleated, and where necessary crumpled inward, so that it came to occupy the interior of the box the three faces now defined. This material was not trimmed or discarded at any point. It was compressed and redirected until it fit inside the boundary that the three marked faces had established, becoming the hidden bulk of the finished object.

The photographs of this sequence show exactly this progression: a flat sheet with a crease pattern and blue markings visible on regions destined to become exterior faces; a partially folded intermediate state in which those regions are beginning to approach one another; a recognizable, if visibly imperfect, cube with the three marked squares legible as its faces; and, in a separate but related sequence, the same sheet carried further, through a series of increasingly dense crumples, into shapes that are no longer cube-like at all, while the same marked regions remain identifiable within them. This second sequence is discussed at length in later chapters, since it demonstrates that the same marked sheet admits more than one admissible resolution and that the cube was one outcome among a family of outcomes compatible with the same declared constraints.

1.5 What was, and was not, discarded

The distinction between the three selected faces and everything else deserves to be stated as sharply as possible, because the entire argument of the monograph depends on it.

The unselected material was not waste. It was not treated as scrap to be trimmed away once its purpose, holding the three squares in their original flat positions, had been served.

It remained physically present throughout, attached to all three faces at every stage, and it remains present inside the finished object. What changed was not its existence but its visibility: it went from being spread out and legible on the flat sheet to being folded, layered, and hidden inside a boundary that no longer displays it.

This is the first appearance of a distinction that recurs throughout the monograph, between a substrate, a realized exterior, and a hidden remainder. The substrate is the whole sheet. The realized exterior is the small fraction of it, in this case three squares, that was selected in advance and that remains legible in the finished object. The hidden remainder is everything else: present, load-bearing in the sense that it gives the object its bulk and its stability, but not designed to be looked at.

1.6 What the experiment demonstrates, precisely

It is important to state the result of this experiment narrowly, because it is tempting to overstate it.

The Crinkle-Cut Supercube does not demonstrate that any assignment of three non-contiguous squares to face roles on any sheet can be folded into a coherent cube corner. It demonstrates one instance. What can be said with confidence is this:

> *At least one non-contiguous, locally oriented face assignment can be realized as a coherent box boundary when the intervening surface is permitted to infold.*

This is a constructive existence claim, and its force comes entirely from the fact that it is constructive. No argument by symmetry, no appeal to general folding theorems, and no computational search established that this configuration was achievable. A physical sheet, folded by hand, established it, by actually reaching the required configuration. The object itself is the proof, in the sense that its existence settles the only question that was actually at issue: whether the declared relations among the three marked squares were realizable at all, given a sheet large enough and permission to infold everything else.

What the experiment leaves open, and what the rest of the monograph is largely occupied with, is how far this single instance generalizes. Does the same strategy work for four squares instead of three? For different relative placements, different orientations, different ratios of selected area to total sheet area? Does it work reliably, or did this particular configuration happen to be an easy one? These are the questions that motivate the formal apparatus developed in Part II, and they cannot be answered by staring harder at the single artifact described in this chapter. They require a vocabulary in which the artifact's construction can be described precisely enough to be varied, tested, and reasoned about beyond the one sheet that was actually folded.

Chapter 2 develops the reason such a vocabulary does not already exist within conventional origami description, and identifies exactly where the descriptive gap lies.

Chapter 2

Why Existing Origami Notation Is Insufficient

2.1 What a fold diagram is for

Origami diagramming, in the form most widely used today, descends from the notation system introduced by Akira Yoshizawa and later standardized and extended by Samuel Randlett and Robert Harbin. Its vocabulary is by now familiar even to casual folders: a dashed line for a valley fold, a dash-dot line for a mountain fold, arrows indicating the direction and sometimes the destination of a fold, and further symbols for operations such as reversing a fold, sinking a point inward, rotating the model, or inflating a pocket.

This notation answers a specific question, asked at a specific moment in a construction: what should the folder do next? It is addressed to a reader who has the model in hand, at a known intermediate state, and who needs to know which edge to bring to which, in which direction, before turning the page. The notation succeeds by being unambiguous at exactly this local level. It does not need to explain why the fold is being made, what larger structure it contributes to, or what would happen if a different fold were made instead. It is an instruction, not an explanation, and it is a good one precisely because instructions of this kind are what a folder executing a known design actually needs.

2.2 What a crease pattern is for

A crease pattern generalizes this a step further. Rather than a sequence of individual instructions, it presents the complete set of fold lines, mountain and valley, superimposed on the flattened sheet, from which the entire model can in principle be folded at once, provided the folder can determine a workable order for collapsing the pattern.

Crease patterns support a substantial body of mathematics: theorems about which patterns can be folded flat at all, results connecting local vertex configurations to global flat-foldability, techniques for designing a crease pattern to produce a specified silhouette, and methods for verifying that a proposed folding sequence does not require the paper to pass through itself. This mathematics is powerful, and it answers a different question than a step-by-step diagram does: not what should the folder do next, but what is the complete geometric relationship between this flat sheet and this folded form?

Both notations, the sequential diagram and the global crease pattern, share an assumption that is worth making explicit because the Crinkle-Cut Supercube violates it. Both assume that the relevant crease structure, whether presented incrementally or all at once, is knowable and statable in advance, and that describing the fold is largely equivalent to describing the object. Once the creases are given, the object is determined, up to the ordinary variability of hand execution.

2.3 Practices that already relax this assumption

This assumption has already been relaxed in several existing origami practices, and it is worth being clear about how far those practices go, so that what remains undescribed after them can be located precisely.

Wet folding introduces controlled curvature into what would otherwise be flat facets, using dampened paper's plasticity to hold gentle curves rather than sharp creases. The resulting surface is not fully specified by a crease pattern in the traditional sense, since the exact curvature achieved depends on how the paper is shaped by hand while damp, but the target form is still generally known in advance and the folder is still working toward a predetermined shape.

Curved folding extends this further, using continuous curved creases whose interaction produces developable surfaces that are difficult to specify by discrete fold lines alone. This mathematics is more demanding than flat-crease theory, but it remains a theory of how a known target surface arises from a known family of creases.

Crumpling, treated as a technique in its own right rather than as an accident to be avoided, appears in sculptural and textural origami, where an artist may crumple paper deliberately to produce organic, non-planar surface texture as part of a larger design. Here the exact microstructure of the crumple is genuinely left unspecified. But the crumpled region is typically decorative or textural. It is not usually required to satisfy a precise exterior relation to some other, separately specified part of the model. Its underdetermination is a feature of its appearance, not a device for realizing a boundary constraint that would otherwise be unreachable.

Each of these practices weakens the connection between crease specification and final form in a particular, limited way. None of them provides a vocabulary for the situation the Supercube presents: several widely separated regions of a sheet, each carrying its own oriented target relation to the finished exterior, with the entire remainder of the sheet left free to do whatever is necessary, structurally rather than decoratively, to bring those regions into their required relation.

2.4 The question none of these notations ask

It is possible to state the gap directly. Sequential diagrams answer: what should the folder do next? Crease patterns answer: along which lines must the sheet fold to produce this form? Wet folding and curved folding answer variants of the same question, extended to continuous curvature. Textural crumpling answers: how can this region's appearance be varied without a fixed target? None of them answers the question the Supercube's construction was actually organized around:

> *Which relations must hold at the final boundary, regardless of the exact crease network through which they are realized?*

This is a question about the object's exterior, posed independently of any particular route to achieving it. It does not ask which creases to make. It asks what must be true when the folding is finished, leaving the folding itself, and everything about it that does not bear on that final truth, unconstrained. A crease pattern can be seen as one particular, fully-specified answer to a question of this kind, in the special case where every degree of freedom has been resolved in advance. The Supercube's specification is a much sparser answer to the same general kind of question, one that resolves only the exterior relations among three regions and leaves everything else open.

2.5 Four descriptive levels

It is useful to organize this discussion around four distinct levels at which a paper construction can be described, moving from the most local and procedural to the most global and declarative.

At the **operation level**, description consists of individual actions: fold this edge to that one, reverse this flap, sink this point. This is the level of the sequential diagram. It is addressed to the hands, one instruction at a time.

At the **crease level**, description consists of the complete pattern of fold lines on the flattened sheet, without necessarily specifying an order of execution. This is the level of crease-pattern mathematics. It is addressed to the sheet as a whole, but still in terms of where it must fold.

At the **shape level**, description consists of the target three-dimensional form itself: a cube, a crane, a tessellated surface. This is the level at which a finished model is usually named and recognized. It says what the object is, without saying how it got that way.

At the **constraint level**, description consists of a set of relations that selected features of the finished object must satisfy: this edge here, this corner there, these three regions mutually adjacent, this much material relegated to the interior. This level does not specify operations, does not specify a complete crease pattern, and does not even specify the complete shape. It specifies only the subset of the shape's structure that has been chosen to matter, and it leaves everything else, including the exact crease network by which those chosen relations come to be satisfied, as free variables to be resolved during construction rather than before it.

Existing origami description operates almost entirely at the first two levels, with the third level, the target shape, usually serving as an informal label attached to a complete crease-level specification rather than as an independent description in its own right. The Crinkle-Cut Supercube's specification belongs to the fourth level. Three squares and their edge and corner annotations constitute a constraint-level description: sparse, declarative, and silent about the crease network required to satisfy it. The remaining chapters of Part II are devoted to giving this fourth level a formal vocabulary of its own, since none of the existing levels can be stretched to cover it without losing the very property, the deliberate underdetermination of everything not explicitly marked, that makes the Supercube's construction what it is.

2.6 Reproducibility reconsidered

One further consequence of working at the constraint level is worth drawing out before moving on, because it changes what it would mean for someone else to reproduce the Crinkle-Cut Supercube.

A conventional diagram aims at a strong form of reproducibility: a second folder, following the same instructions, should arrive at very nearly the same crease network and the same final layering as the original. Fidelity to the instructions is fidelity to the object.

A constraint-level specification supports, and arguably only supports, a weaker but more interesting form of reproducibility. A second folder given the same three marked squares and the same edge and corner annotations, but no information about how the original sheet was actually folded, might arrive at a finished cube corner by an entirely different sequence of folds, with an entirely different interior crease network and layering, while still satisfying the same declared relations. Both constructions would be, in a sense to be made precise in Chapter 6, equivalent: not because they share a folding history, but because they share a boundary.

This is not a weaker notion of success. It is a different one, and it is the notion the rest of this monograph will be built around. Two realizations count as satisfying the same specification when they agree on what was declared to matter, whatever else may differ between them. Chapter 3 begins the work of stating what, exactly, is declared to matter, and how.

Part II

A Formal Language of Infolding

Chapter 3

Marked Surfaces and Local Frames

3.1 The sheet as a material surface

The formal apparatus developed in this and the following two chapters is built to describe exactly the situation Chapter 1 reconstructed: a single connected sheet, most of it destined to disappear from view, a small number of regions on it singled out in advance, and a target relation among those regions that the finished object must satisfy.

Let P denote the sheet. At the coarsest level of description, P can be treated as a connected, bounded two-dimensional surface, initially lying flat in the plane. This idealization is useful for stating relations precisely, but it should not be mistaken for a claim that the sheet's material properties are irrelevant. Real paper has thickness, stiffness, grain direction, and a limited capacity to bend before it creases plastically rather than elastically; it resists self-intersection and cannot pass through itself. These properties govern what is physically achievable and are reintroduced wherever the argument depends on them, particularly in Chapter 6's treatment of surplus area and Chapter 9's treatment of obstructions. For the purpose of stating what is being specified, however, it is enough to treat P as a surface whose points can be tracked through a deformation.

3.2 Selected regions and the infoldable remainder

A finite collection of regions is selected on P in advance:

$$E = \{E_1, E_2, \ldots, E_n\}$$

These are the regions intended to remain legible on the finished object's exterior. In the Supercube, $n = 3$, and each E_i is one of the three marked squares.

The complement of this selection,

$$I = P \setminus E$$

is the infoldable remainder: every part of the sheet not singled out for exterior legibility.

The word "infoldable" is chosen deliberately over alternatives like "excess" or "waste," because those alternatives suggest that I is material to be gotten rid of, and it is not. Nothing about I is discarded, cut, or treated as disposable. What distinguishes I from E is only that its final geometry has not been declared in advance. It is free where E is fixed. This is a statement about how much has been specified, not a statement about the region's importance to the finished object; as Chapter 6 will argue at length, I typically constitutes the great majority of the object's material bulk even though it constitutes none of its declared exterior.

3.3 Local frames

Each selected region carries more than just its own extent. It carries an orientation: a declaration of which of its features correspond to which roles in the target exterior.

Write a framed region as a pair:

$$F_i = (E_i, o_i)$$

where o_i is the orientation data attached to E_i. For a square region, this orientation data might distinguish a bottom edge from a side edge from a top edge, or equivalently specify a distinguished corner together with the two edge directions incident to it. The precise content of o_i depends on the shape of E_i and on how many of its features are given a target role; nothing in the formalism requires E_i to be a square, or requires every edge of E_i to be labeled.

In the Supercube, the physical blue marks are best understood as a direct, material instantiation of this orientation data. They were not drawn to indicate where the sheet should fold. They were drawn to indicate which edge of each marked square was to become which edge of the finished cube, independent of the square's position on the original flat sheet or its distance from the other two marked squares. A local frame, in other words, is portable in exactly the sense the marked squares needed to be: it travels with the region through whatever deformation eventually brings that region into its target position, and it is legible on the region wherever that region ends up.

3.4 What a local frame does not specify

It is worth being precise about the boundary of what a local frame declares, because the formalism depends on this boundary being sharp.

A local frame $F_i = (E_i, o_i)$ specifies the identity of a region and the roles assigned to some of its features. It does not specify where E_i currently sits relative to the other selected regions. It does not specify how E_i is to be moved, folded, or reoriented in order to bring o_i into agreement with the corresponding roles on the target object. And critically, it says nothing at all about the material outside E_i, including the material immediately adjacent to it. A local frame is a statement about one region, made independently of every other region and independently of the deformation that will eventually be required to realize it.

This independence is what makes it possible to declare relations among widely separated regions without having to say anything about the intervening sheet. The three squares of the Supercube could be marked without reference to one another and without reference to the material lying between them, because each mark was a self-contained statement about one region's intended role. The relations *among* the three frames, which edges must meet which, which corners must coincide, are a separate layer of specification, introduced in Chapter 4. A local frame, on its own, only ever describes a single region's target orientation.

3.5 Edge roles and partial labeling

Within a local frame, individual edges (or, more generally, individual boundary features) can be assigned symbolic roles. A short, non-exhaustive vocabulary suffices for most purposes:

- B, for an edge intended to become a bottom edge of the target object;
- S, for an edge intended to become a side edge;
- T, for an edge intended to become a top edge;
- J_k, for an edge intended to join a specific other labeled edge, indexed by k;
- V_k, for an edge incident to a designated target vertex, indexed by k;
- X, for an edge explicitly prohibited from exterior exposure, should such a prohibition be needed.

A framed region can then be written compactly. For instance, one of the Supercube's three squares might be written:

$$F_1 = [E_1;\ B = e_1,\ S = e_2]$$

indicating that edge e_1 of region E_1 is assigned the bottom role and edge e_2 is assigned the side role, with the remaining edges of E_1 left unlabeled.

That last clause, the remaining edges left unlabeled, is not an oversight to be corrected in a more careful specification. It is the specification. A local frame need not assign a role to every edge of its region. Leaving an edge unlabeled is a positive declaration that the final orientation or adjacency of that edge is not being constrained, not a gap that a more thorough author would have filled in. This is what is meant, throughout this monograph, by a **partial specification**: not an incomplete draft of a complete one, but a deliberately incomplete final statement of exactly those relations the constructor has chosen to care about.

3.6 Why partiality is load-bearing

It is worth pausing on why this partiality matters, rather than treating it as an incidental economy of notation.

A specification that labeled every edge of every region, and additionally specified the exact relation of every point of I to every other point, would no longer be a constraint-level description in the sense of Chapter 2. It would have collapsed back into something closer to a complete crease pattern: a full account of the finished object's geometry, articulated region by region rather than crease by crease, but no less total in its coverage. Total specification of this kind is not wrong, but it is a different project. It reintroduces exactly the assumption Chapter 2 identified as the limiting feature of existing notation, that describing the object is equivalent to describing its complete geometry, and it forecloses the very freedom that allowed the Supercube's surrounding material to find its own accommodation.

The formalism's usefulness depends on keeping the labeled portion small relative to the unlabeled portion. A specification that declares three edges and one shared vertex, out of a sheet with area many times larger than the three regions those edges belong to, constrains a correspondingly enormous space of possible deformations down to whatever subset of that space happens to satisfy the three declared relations. The interesting content of the theory, developed across the remaining chapters of Part II and the whole of Part III, concerns what can be said about that subset, how large or small it is, what its members have in common beyond the declared relations themselves, and what physical process might be relied upon to locate a member of it. None of this content is visible if the specification is allowed to grow until it covers the whole sheet.

3.7 Sparse specification as the chapter's central idea

The formal objects introduced in this chapter, the substrate P, the selected regions E and their complement I, and the local frames F_i with their partially labeled edge roles, are deliberately minimal. They say only as much as the Supercube's actual construction said: which regions matter, and which of their features are assigned target roles. Everything else about the sheet's eventual geometry is left open by construction, not by omission.

This is the sense in which the Supercube's specification is **sparse**: a small, explicitly bounded set of declared relations governing a much larger surface, with no claim made, and no claim needed, about anything the specification does not mention. Chapter 4 extends this vocabulary from single regions to relations among regions, stating formally what it means for two or more framed regions to be required to meet, align, or coincide at the finished object's boundary, and introduces the graphic notation by which such a specification can be recorded on the sheet itself, in the same spirit as the original blue marks.

Chapter 4

Boundary Grammar and Infolding Notation

4.1 From single frames to relations among frames

Chapter 3 gave a vocabulary for describing one selected region at a time: its identity, its extent, and the roles assigned to some of its edges. That vocabulary is not yet sufficient to state the Supercube's specification, because the specification's real content lies in the relations *among* the three marked squares, not in any one square considered alone. This chapter introduces that relational layer, together with a compact graphic notation for recording it directly on a sheet.

Let Ω denote the effective volume of the object that results from a given deformation, and let $\partial\Omega$ denote its apparent exterior boundary. A **realization** is a deformation f carrying the marked sheet P into a folded state that satisfies a declared set of constraints C. The remainder of this chapter defines the constraint types that make up C.

4.2 Exposure

The simplest constraint asserts that a selected region ends up visible on the finished exterior:

$$f(E_i) \subset \partial\Omega$$

within whatever tolerance the construction's purpose demands. An exact reading of this condition would require every point of E_i to lie precisely on the boundary surface; a more forgiving reading, appropriate to a hand-folded object like the Supercube, allows E_i to be substantially, though not perfectly, exposed, with minor overlap or partial occlusion at its edges tolerated. Chapter 6 returns to the question of how such tolerance should be measured; for now it is enough to note that an exposure condition is a claim about where a region ends up, not about its exact shape once it gets there.

4.3 Adjacency

An adjacency condition requires a specified edge of one region to meet a specified edge of another in the finished exterior:

$$A(E_i : e_a,\ E_j : e_b)$$

This notation reads: edge e_a of region E_i must be adjacent, in the realized object, to edge e_b of region E_j. Adjacency can be directed, when the orientation of the meeting matters, distinguishing an aligned meeting from a reversed one, and it can further distinguish a flush meeting from a merely incident one, in which the two edges touch without running parallel along their full length. The Supercube's construction relied on adjacency conditions of exactly this kind to specify that the bottom edge of one marked square was to meet the side edge of another.

4.4 Vertex-gluing

Where three or more frames are required to meet not merely edge-to-edge but at a single common point, a vertex-gluing condition is used:

$$G(F_1, F_2, F_3) = V$$

This states that the designated corners of the three named frames must become incident to one shared exterior vertex V in the realized object. This was the single most consequential condition in the Supercube's specification. Chapter 1 described the moment at which this condition first became satisfied as the decisive turn in the construction: once the three marked corners could be brought together at one point with their associated edges correctly oriented, the object possessed a coherent partial cube frame, even though the great majority of the sheet remained unresolved. A vertex-gluing condition is, in this sense, disproportionately powerful relative to its brevity: it ties together conditions on several frames at once, and its satisfaction tends to lock a large amount of the surrounding geometry into place as a side effect.

4.5 Orientation

A region may satisfy every adjacency and vertex condition demanded of it while still being incorrectly rotated or reflected relative to its intended role, a bottom edge occupying a side position, for instance, even where the correct edges happen to touch. An orientation condition rules this out by requiring the realized deformation to carry a frame's declared orientation into agreement with its target orientation:

$$Df_i(o_i) \approx O_i$$

The notation here is not meant to invoke a literal derivative or to require smooth differentiability of f everywhere; the sheet's actual deformation includes sharp creases at which no such derivative would be classically defined. What the notation expresses is the preservation of the directional roles declared in o_i: that whatever the deformation does elsewhere, the marked bottom edge ends up oriented as a bottom edge, the marked side edge as a side edge, and so on, up to the tolerance the construction allows.

4.6 Relegation

The infoldable remainder I, defined in Chapter 3, is not left entirely unconstrained. It is subject to one standing condition, which can be stated as a relegation constraint:

$$f(I) \subset \Omega \cup H$$

where H denotes the collection of hidden boundary layers, seams, overlaps, and interior pockets the realized object may contain. This is not a claim that the free material occupies a literal, zero-thickness mathematical interior; a folded sheet has thickness, and its layers press against one another and against the interior surfaces of the exposed faces. The condition should be read practically: the free material must end up tucked inside the apparent envelope of the object, or hidden beneath its designated exterior faces, rather than protruding as unaccounted-for exterior surface of its own. Relegation is what allows the object's declared exterior, in the Supercube's case just three squares, to remain the only legible surface of an object built from a sheet many times larger than those three squares combined.

4.7 Closure

A final constraint type governs how complete, stable, or functionally container-like the finished object must be. Closure is deliberately left as a family of conditions rather than a single formula, since what counts as adequate closure depends entirely on the construction's purpose. Possible closure conditions include: that the boundary contains no large unintended openings; that the relegated material remains contained without external support once the folder's hands are removed; that the object holds its shape for some specified duration; or, at the least demanding end, only that the vertex-gluing and adjacency conditions above remain satisfied, with no further requirement on the object's overall rigidity or permanence.

The Supercube's own closure was modest by this reckoning: the three marked faces held their adjacency and vertex relations, and the bulk of the sheet was contained within the resulting boundary, but the object was not rigid, was not sealed against all gaps, and would not have survived indefinite handling without some of its relations loosening. This is treated in this monograph not as a shortfall but as an accurate report of exactly which conditions the construction was actually able to secure, a report that later chapters, particularly Chapter 9's discussion of protected and sacrificial distinctions, will show was itself a meaningful outcome rather than a simple failure to finish the job.

4.8 The boundary grammar as a whole

Together, exposure, adjacency, vertex-gluing, orientation, relegation, and closure constitute what this monograph calls a **boundary grammar**: a set of relation types by which a finished object's exterior can be specified without specifying the deformation that produces it. The Supercube's full specification can now be written compactly, gathering the pieces introduced across this chapter and the last:

$$\begin{gathered}\mathrm{INFOLD}(P \setminus E \mid \mathrm{Exposure}(E),\ \mathrm{Adjacency}(A),\\ \mathrm{Orientation}(O),\ \mathrm{Vertex}(V),\ \mathrm{Closure}(K))\end{gathered}$$

The left-hand argument, $P \setminus E$, names the free material to be searched over. The right-hand arguments name the boundary grammar conditions that free material's eventual configuration must satisfy, or must not prevent the rest of the object from satisfying. Chapter 5 takes up the INFOLD operator itself, and the search process it denotes, in detail. The remainder of this chapter turns from the algebraic notation above to a graphic notation suited to marking these same conditions directly on a physical sheet, in the spirit of the original blue annotations.

4.9 A graphic vocabulary

A workable notation for constraint-directed infolding should be usable by hand, without requiring a computational model to interpret it, and it should visually distinguish protected material from free material at a glance. The following elements are proposed to meet those requirements; the exact graphic conventions are not claimed to be uniquely correct, only sufficient, and Appendix A collects them in a single reference sheet.

Protected faces. A selected region E_i is marked with a heavy outline and given a short identifier, F_1, F_2, F_3, and so on. The heavy outline distinguishes it visually, at a glance, from the surrounding free material.

Free material. Regions belonging to I, the infoldable remainder, are left unmarked or given only a light shading. This absence of marking is itself informative: it signals that the region's internal crease structure is outside the specification, not merely that it has not yet been decided how to draw it.

Edge-role ticks. Rather than adopting mountain- and valley-fold lines for this purpose, which already carry a specific and different meaning, edge roles are marked with short ticks or letters placed near the relevant edge: a single solid tick for a bottom-role edge, a double tick for a side-role edge, and a matching pair of letters, placed near two edges required to meet, to indicate an adjacency condition between them.

Vertex labels. A small circled label, such as V_1, repeated at a corner of each of several selected regions, indicates that those corners are required to become incident to one shared exterior vertex. Repetition of the same label across multiple regions is what carries the vertex-gluing condition; a single unrepeated label carries no such requirement.

Exposure status. Where a region's exposure is not simply assumed, a short status label may be attached: OUT for a region intended to be principally exterior, IN for one intended to be principally interior, FREE where either status is acceptable, and SEAM for a region permitted to appear only as a narrow boundary strip or overlap rather than as a full exterior face.

Phase annotations. Where a construction benefits from being described in stages, brief phase labels may be added separately from the geometric marks themselves: *realize* V_1, then *align the bottom-edge cycle*, then *relegate the remaining surplus.* These annotations describe the order in which constraints are to be secured, not the exact hand motions by which any one of them is secured; they are closer to a solver's schedule than to a fold-by-fold diagram.

4.10 Compatibility with conventional symbols

None of the notation introduced in this chapter is intended to replace the existing vocabulary of mountain and valley folds, reversals, sinks, and the rest. A hybrid diagram is not only permitted but often desirable: conventional fold symbols can specify a known local operation, perhaps a reversal used to bring two marked edges into contact, while the constraint-level symbols introduced here specify what that operation is being performed in order to preserve or achieve. The two vocabularies operate at different descriptive levels, in the sense of Chapter 2, and a complete practical diagram for a constraint-directed construction may well need both: the constraint level to state what must be true when the folder is finished, and the operation level to record whatever specific moves the folder actually found useful along the way.

Chapter 5 turns to the process this notation describes rather than merely records: the search, denoted INFOLD, by which a folder actually locates a deformation satisfying a declared boundary grammar, and the auxiliary operations, locking, relaxing, and repairing a constraint mid-construction, that make such a search tractable by hand.

Chapter 5

The Infold Operator and Constraint-Preserving Search

5.1 A different kind of instruction

The notation developed in Chapter 4 describes what a finished realization must satisfy. It does not, by itself, describe the process of arriving at one. This chapter introduces that process, in the form of a single operator that stands in deliberate contrast to an ordinary fold instruction.

An ordinary fold instruction, of the kind Chapter 2 associated with the operation level of description, names a specific action: bring this edge to that one, in this direction. Executed correctly, it produces one determinate result. The operator introduced here, written INFOLD, does not name a specific action and does not produce one determinate result. It names a family of permitted actions and a set of conditions those actions must eventually satisfy, and it leaves the choice of any particular sequence of actions unresolved.

$$\text{INFOLD}(I \mid C)$$

reads: deform the free region I through any combination of the permitted operations, in any order and to any extent, such that the constraint set C comes to be satisfied, or is not thereby made unsatisfiable. INFOLD is better understood as a search instruction than as a fold instruction. It tells the constructor what must be true at the end without telling them how to get there, in the same sense that a specification tells an engineer what a bridge must bear without telling them which member to weld first.

5.2 The Supercube's instruction, written out

Gathering the constraint types introduced in Chapter 4, the Supercube's full specification can be written:

$$\text{INFOLD}(P \setminus E \mid \text{Exposure}(E),\ \text{Adjacency}(A),$$
$$\text{Orientation}(O),\ \text{Vertex}(V),\ \text{Closure}(K))$$

Read in full: deform the material outside the three marked squares, by any combination of folding, pleating, and crumpling, such that the three squares remain exposed, their marked edges meet as declared, their marked orientations are preserved, their marked corners coincide at a shared vertex, and the resulting object achieves at least the modest closure the construction actually required. Nothing in this instruction specifies which fold to make first, how many creases the free material should acquire, or what shape that material takes once it is no longer visible. Those questions are left to whatever process actually carries out the search, which is the subject of the remainder of this chapter and, from a different angle, of Chapter 7.

5.3 The permitted operation family

INFOLD ranges over a specific, if broad, family of physical operations. Bending, buckling, pleating, corrugating, sinking, local inversion, crumpling, nesting, overlapping, wrapping, and the insertion of one layer beneath another are all permitted. Cutting and adhesive bonding are not permitted by default; a construction that requires either should say so explicitly, since both operations change the sheet's connectivity in ways the rest of this monograph's argument, particularly Chapter 6's treatment of intrinsic connectivity, depends on being ruled out unless stated otherwise. Tearing occupies an intermediate position: depending on the construction's purpose, an accidental tear may either invalidate the realization outright or be treated as a separate, weaker category of success, a distinction Chapter 9 returns to when it discusses obstructions to realizability.

5.4 Protected and sacrificial geometry

The asymmetry between the three marked squares and the rest of the sheet, already familiar from Chapters 3 and 4, can now be stated as an asymmetry in how much freedom INFOLD grants to different parts of I.

Some geometry is **protected**: it corresponds directly to a declared constraint, and INFOLD must not sacrifice it in the course of resolving anything else. The three marked squares' exposure, their edge roles, and their shared vertex are protected in this sense throughout the Supercube's construction.

Other geometry is **sacrificial**: its precise final shape and its crease history are explicitly not protected, and INFOLD is free to let this geometry take whatever form is convenient for satisfying the protected constraints. It is important to be precise about what "sacrificial" means here, since the word can misleadingly suggest destruction. No material is destroyed. What is sacrificed is specificity: the constructor gives up any claim to control the exact shape this geometry ends up taking, in exchange for the freedom that surrender buys elsewhere in the construction. The bulk of I in the Supercube is sacrificial in exactly this sense. Its final crumpled form was never specified, was not predictable in advance, and does not need to be reproduced by a second constructor working from the same specification; only the protected relations need to be reproduced.

This asymmetry, high constraint on a small selected exterior, low constraint on a large hidden connective mass, is the operational heart of constraint-directed infolding. It is what allows a sparse specification, in the sense of Chapter 3, to be realized at all: the freedom given up on the sacrificial majority of the sheet is exactly the freedom that makes it possible to satisfy demanding relations on the small protected minority.

5.5 Locking a relation once achieved

A search process that has to keep every constraint simultaneously at risk throughout construction is harder to carry out, by hand or otherwise, than one that can secure a relation and then set it aside. The Supercube's own construction, as reconstructed in Chapter 1, relied on securing the vertex-gluing condition first and only then addressing the remaining material, and this reliance can be given a name.

$$\mathrm{LOCK}(C_k)$$

denotes the decision to treat a currently satisfied constraint C_k as provisionally fixed, so that subsequent operations are chosen, or avoided, specifically to preserve it. For the Supercube, once the three marked corners had been brought together at a shared vertex, that relation was locked:

$$\mathrm{LOCK}\big(G(F_1, F_2, F_3) = V\big)$$

and the remaining infolding was carried out under this lock, meaning that no subsequent manipulation of the free material was permitted to disturb the vertex once it had been achieved. Combined with the operator introduced above, the full sequence can be written:

$$\mathrm{INFOLD}\big(I \mid \mathrm{LOCK}(V),\ \mathrm{Closure}(K)\big)$$

This expresses precisely the strategy Chapter 1 described in narrative form: establish the three-face corner first, treat it thereafter as fixed, and resolve everything else in whatever way that fixed corner permits.

5.6 Relaxing a lock that has become an obstruction

A locked relation, useful as it is for organizing a search, can also become an obstruction. Material rearranged elsewhere in the sheet may, at some point, be unable to proceed without disturbing a relation that was locked earlier. Rather than treating this as a dead end, the formalism permits a controlled relaxation:

$$\mathrm{RELAX}(C_k, \varepsilon)$$

which permits the previously locked relation C_k to drift, temporarily, within a stated tolerance ε, so that the surrounding obstruction can be cleared. This is not an abandonment of the constraint. It is an explicit, bounded loosening, made in service of eventually satisfying the construction as a whole, and it is paired with a corresponding operation for restoring the relation once the obstruction has passed:

$$\mathrm{REPAIR}(C_k)$$

which reasserts C_k, ideally within its original tolerance, once the surrounding material has been rearranged sufficiently to permit this.

5.7 Why relaxation and repair matter beyond bookkeeping

It would be possible to regard LOCK, RELAX, and REPAIR as mere bookkeeping devices, convenient shorthand for describing a process that could equally well be described without them. This understates their significance.

What these three operations jointly capture is that constraint-directed infolding, as actually practiced by hand, is not a monotonic process in which each constraint, once satisfied, remains satisfied until the end. It is a process with local setbacks that are recoverable precisely because the constraint was never irreversibly fused with the surrounding sheet; a locked relation, expressed this way, is always understood to be provisionally locked, subject to exactly the kind of temporary loosening RELAX names. A notation that only recorded successes, and had no vocabulary for a temporary, bounded, intentional step backward, would misrepresent how the construction actually proceeded, and would offer no way to distinguish a principled, recoverable relaxation from an accidental, possibly unrecoverable failure of the same relation. Chapter 7 develops the embodied side of this same point, treating a blocked fold not as an error external to the search but as information about the current boundary of what is reachable; RELAX and REPAIR are this chapter's formal counterpart to that later, more experiential discussion.

5.8 What Chapter 5 has and has not established

This chapter has given the Supercube's construction a precise operational vocabulary: an operator, INFOLD, denoting constrained search over a permitted family of deformations; a distinction between protected and sacrificial geometry that explains why such a search is tractable at all; and a small set of auxiliary operations, LOCK, RELAX, and REPAIR, that describe how a constraint can be provisionally secured, temporarily loosened, and subsequently restored over the course of a single construction.

What it has not established, and what the remaining chapters of the monograph take up in turn, is any general account of when a given INFOLD instruction is satisfiable at all, how large the space of satisfying deformations tends to be, or what, if anything, the members of that space have in common beyond satisfying the same declared constraints. These are the questions Part III addresses, beginning with Chapter 6's treatment of the relationship between a sheet's original connectivity and the exterior adjacency its infolded remainder makes possible.

Part III

Hidden Geometry and Exterior Underdetermination

Chapter 6

Adjacency, Surplus Surface, and Historical Interior

6.1 Two graphs on the same sheet

The Supercube's construction can be described by two different graphs, and the entire theoretical argument of this chapter follows from noticing that these two graphs need not agree.

The first is the **sheet graph**, written G_0, whose nodes correspond to the marked regions and the intervening portions of the flat, unfolded sheet, and whose edges record which of these regions are directly connected across the original planar surface. In the Supercube, G_0 records the fact that the three marked squares are widely separated, joined only through a large connected mass of unmarked material, with no two of the three squares sharing an edge anywhere on the flat sheet.

The second is the **boundary graph**, written G_∂, whose nodes correspond to the regions actually visible on the finished exterior, and whose edges record which of these regions are adjacent to one another in the realized object. In the Supercube, G_∂ records that the three marked squares, after folding, are mutually adjacent, meeting pairwise along the edges their annotations declared and sharing the common vertex their vertex-gluing condition demanded.

The Supercube's construction was organized around producing a specific, wanted disagreement between these two graphs:

$$G_\partial \neq (\text{the exposed subgraph induced by } G_0)$$

The three squares are not adjacent in G_0. They are adjacent in G_∂. Something has to account for this difference, and it is worth being precise about what does and does not account for it.

6.2 What changes, and what does not

It might seem that producing new adjacency where none existed before requires cutting the sheet, since cutting is the ordinary way material connectivity is altered. This is not what happened, and the reason it is not what happened is the central fact this chapter exists to draw out.

The sheet's intrinsic material connectivity, its status as a single connected piece of paper with no tears or seams introduced, is unchanged by the construction. Every point of the original sheet remains connected to every other point by some continuous path through the material, exactly as it was before any fold was made. What changes is which of these connections lie on the exterior boundary of the finished object and which have been folded away from it.

The two squares that end up adjacent in G_∂ were never made adjacent in G_0; no new material connection was created between them. Instead, the material that used to separate them in the flat sheet, previously visible and lying between them in the plane, has been displaced away from the exterior, folded inward, so that the two squares can now touch directly at the boundary while the displaced material persists, hidden, along a folded and occluded route that still connects them exactly as it always did, just no longer where anyone looking at the finished object can see it.

This can be stated as a general principle, though a modest one: new visible adjacency does not require new material connectivity. What the construction changes is which relations among the sheet's points are expressed at the boundary, not the sheet's underlying connectedness. The apparent paradox, that two regions never touching in the flat layout end up touching in the folded object, dissolves once hidden layers are admitted as a legitimate place for material to be, rather than treated as an embarrassment to be minimized.

6.3 The exterior as a quotient

A useful way to state this relationship formally is to treat the finished, visible object as a coarse quotient of the complete folded state. The complete folded state includes every layer, every fold, every point of contact between one part of the sheet and another, all of it. Most of this detail becomes observationally indistinguishable once only the exterior is inspected: many different interior arrangements of hidden layers would produce the same visible exterior, and an observer confined to looking at the outside of the object has no way to tell which of these arrangements actually occurred.

The quotient retains exactly what the boundary grammar of Chapter 4 declared mattered: face identity, orientation, and the adjacency relations among exposed regions. It suppresses everything else, not because everything else is unimportant to the object's physical structure, it plainly is important, load-bearing, and literally what gives the object its bulk, but because the exterior, considered purely as a visible surface, does not preserve enough information to reconstruct it.

6.4 Surplus area

The preceding two sections established that new exterior adjacency can be produced without new material connectivity, provided there is somewhere for the intervening material to go. This section asks how much material that actually involves, since the answer bears directly on how demanding a given specification turns out to be in practice.

Define the **surplus ratio**:

$$\sigma = \frac{\text{Area}(P) - \text{Area}(E_{\text{visible}})}{\text{Area}(E_{\text{visible}})}$$

where $\text{Area}(P)$ is the total area of the original sheet and $\text{Area}(E_{\text{visible}})$ is the combined area of the regions actually intended to remain exposed. A conventional cube net, built from close to the minimal area six unit squares require, has a surplus ratio near zero: essentially all of the sheet becomes visible exterior, give or take whatever small tabs are needed for joining. The Supercube's surplus ratio was, by contrast, substantial: the three marked squares accounted for only a modest fraction of the sheet's total area, with the remainder, several times larger, constituting pure surplus, none of it intended for exterior legibility.

A high value of σ is not, by itself, a difficulty. It is a resource. The surplus material is exactly what makes the vertex-gluing and adjacency conditions of Chapter 4 achievable in the first place, since it is the material that gets folded away to create the separation between the flat layout and the folded exterior that Section 6.2 described. What a high surplus ratio does introduce is a downstream question: where does all of that material actually go once it has been folded away, and what does accommodating it cost the rest of the construction. That question is taken up directly in Section 6.5.

6.5 The central tradeoff

The material displaced by a high surplus ratio does not vanish; it has to be accommodated somewhere inside the boundary the exposed faces define, and accommodating it is not free. This yields the tradeoff at the center of this chapter's argument:

$$\text{exterior regularity} \leftrightarrow \text{hidden interior complexity}$$

A cleaner, more regular exterior, flatter faces, sharper edges, more exact adjacency, does not come for free when the surplus ratio is high. It is typically purchased by a more complicated interior: a greater number of hidden layers, sharper curvature concentrated out of view, more stored elastic energy in the tightly folded material, or a more elaborate route by which the surplus is threaded through the available interior space without disturbing the exposed faces it is tucked behind.

This tradeoff is visible directly in the photographic record of the Supercube's construction. The version of the object in which the three marked squares present as relatively clean, recognizable faces required the surrounding material to be folded and crumpled quite densely to fit inside; the alternative crumpled states, photographed as the same sheet was carried further into increasingly irregular configurations, show the same marked regions still present and still identifiable, but no longer held to the same exterior regularity, with the freed regularity apparently absorbed as a less constrained, more loosely packed interior. Neither state is more or less correct relative to the declared specification. The specification, recall from Chapter 4, said nothing about the sacrificial material's degree of regularity. Both states are realizations of the same boundary grammar, purchased at different points along the same tradeoff.

It is worth naming a related, practical consequence directly: **material congestion**. As more surplus is folded inward, the available interior space fills, and the material already there can begin to resist further rearrangement, potentially forcing an already-established face outward, disturbing an adjacency that had previously been achieved, or blocking a corner from closing correctly. Section 5.6's discussion of relaxation and repair describes the operational response to exactly this kind of congestion; this chapter's contribution is to note that congestion is not an incidental hazard of infolding but a direct, quantifiable consequence of how much surplus a given specification's surplus ratio requires the construction to absorb.

6.6 Creases as a historical record

Every fold introduced during the Supercube's construction left a physical trace on the sheet: a crease, a compressed region of fibers, a line along which the paper's resistance to bending has been permanently reduced. These traces persist even where the fold that produced them is later reversed or relaxed. This section treats them as what they are, a physical record of the construction's actual history, rather than as incidental byproducts of reaching the finished shape.

The declared markings, the three squares and their edge and vertex annotations, state what the finished object was required to achieve. The accumulated crease network states, after the fact, what path the sheet was actually taken along in order to achieve it. These are different kinds of information, and the distinction between them, a declared target and a witnessed trajectory, is one this monograph will return to directly in Section 6.8.

A crease is not merely evidence that a particular fold occurred at a particular moment. It also constrains what can happen afterward. Paper that has already been creased along one line bends more readily along that line in the future and more stiffly elsewhere; a region that has already been compressed into a tight pleat cannot easily be returned to a flat, unstressed state without visible residual curvature. This means that the space of configurations reachable from the sheet's current state depends on the specific history that brought the sheet to that state, not merely on the sheet's current visible shape. Two sheets that look identical at some intermediate stage of construction, but that arrived at that stage by different sequences of folds, may not have the same set of configurations reachable from that point onward.

This dependency is worth naming explicitly:

$$\mathrm{Reach}(P, h_t)$$

denotes the set of configurations reachable from the sheet's current state, given that its deformation history up to time t has been h_t. The notation makes explicit what the previous paragraph argued informally: reachability is a property of the pair, sheet and history, not of the sheet's current shape alone. A construction that has locked a relation, in the sense of Section 5.5, and creased the surrounding material accordingly, has narrowed $\mathrm{Reach}(P, h_t)$ in a way that a differently-creased sheet at the same superficial stage would not have narrowed it.

6.7 Many histories, one boundary

The preceding sections established two facts that combine into this chapter's central result. First, a given exterior specification can be satisfied while displacing surplus material along many different routes through the interior, since the specification says nothing about how that displacement is to be organized. Second, the specific route actually taken is recorded, irreversibly to a first approximation, in the sheet's accumulated crease network, and different routes leave different, generally distinguishable, records.

Putting these together: many different folding histories, producing many different interior crease networks and layer arrangements, can realize the identical exterior boundary specification.

$$\text{many histories} \;\rightarrow\; \text{many interiors} \;\rightarrow\; \text{one boundary class}$$

This is not a hypothetical possibility raised for the sake of argument. It is directly demonstrated by the Supercube's own photographic record: the same marked sheet, carried through different further manipulations, produces both a relatively regular cube-corner exterior and a family of considerably more irregular crumpled exteriors, all of them consistent with the same declared face, edge, and vertex conditions, none of them privileged by the specification over any other. The specification defines a class of acceptable exteriors, or in the strictest cases a class of acceptable exact exteriors together with a much larger class of acceptable approximate ones, and any member of that class, however it was reached, counts as satisfying it.

6.8 Boundary form does not disclose interior history

The consequence of Section 6.7 that matters most for the remainder of this monograph is epistemological rather than constructional, and it deserves to be stated as directly as possible.

Given only the finished, folded exterior of an object realized by constraint-directed infolding, an observer cannot, by inspection of that exterior alone, recover which of the many possible interior histories actually produced it. The exterior does not encode this information, because the specification that governs the exterior was never designed to encode it; recall from Chapter 4 that the boundary grammar constrains only exposure, adjacency, orientation, vertex incidence, relegation, and closure, and says nothing whatsoever about the internal route by which relegation is achieved. An object satisfying that grammar is, from the standpoint of the grammar itself, complete and correct regardless of which member of the many-histories-many-interiors class it happens to be.

This should be distinguished carefully from a claim that the interior is unknowable in principle. It is not unknowable; Chapter 8's discussion of imaging and destructive unfolding takes up exactly how an interior might be recovered by means other than inspecting the exterior. The claim is narrower and, for that reason, sharper: the exterior itself, considered purely as a boundary satisfying a declared grammar, carries no reliable trace of its own history. Recovering that history requires additional access, physical or otherwise, beyond what the finished boundary presents.

This licenses a distinction among several notions of equivalence that have been used loosely up to this point and that now need to be kept separate.

Two realizations are **boundary-equivalent** when they satisfy the same declared exposure, adjacency, orientation, vertex, and closure conditions, regardless of anything else about them. This is the weakest and most important notion for the purposes of this monograph, since it is the notion the Supercube's specification was actually designed to secure.

Two realizations are **structurally equivalent** when they are boundary-equivalent and additionally share the same pattern of interior layer contacts, even if the specific creases producing those contacts differ in detail.

Two realizations are **crease-equivalent** when they share, in addition to structural equivalence, the same crease network up to a specified tolerance, meaning not merely that the same layers touch but that they touch along recognizably the same fold lines.

Two realizations are **procedurally equivalent** when they were produced by the same sequence of operations, in the same order, the strongest and least useful notion of equivalence for present purposes, since it is closest to the strong reproducibility Chapter 2 associated with conventional sequential diagrams.

These four notions form a hierarchy, each one a strictly stronger requirement than the last, and the Supercube's two documented outcomes, the relatively regular cube corner and the family of more irregular crumpled states, are boundary-equivalent to a considerable degree, agreeing on the exposure, adjacency, and vertex conditions the specification actually demanded, while being neither structurally, crease-, nor procedurally equivalent to one another in any strong sense. This is not a defect in either realization. It is the expected consequence of specifying at the constraint level, in the sense of Chapter 2, rather than at the crease or operation level: a constraint-level specification purchases its sparseness at the price of exactly this kind of underdetermination, and the underdetermination is not a loose end to be tied up but the specification doing precisely what it was built to do.

The chapter's argument can now be closed with the proposition it has been building toward throughout:

Exterior equivalence does not imply structural, crease, or procedural equivalence.

A simple, legible exterior can be the visible quotient of a complicated, irreversible, and largely unrecoverable interior process, and nothing about the exterior's simplicity licenses an inference about the simplicity, or even the singularity, of the process that produced it. Chapter 7 takes up the other side of this same fact, asking what it means to conduct the search this chapter has been analyzing after the fact, from the inside, without the benefit of already knowing which interior history one is going to end up leaving behind.

Part IV

Embodied Search

Chapter 7

Folding Without a Complete Internal Image

7.1 The search from the inside

Chapter 6 described the space of possible realizations of a boundary specification from the outside, as an analyst might look back on it once several outcomes were available for comparison. This chapter asks what the same search looks like from the inside, to a constructor who has only the marked sheet, their hands, and the sheet's own resistance to work with, and who does not yet know, and cannot know in advance, which member of the many-histories-many-interiors class described in Section 6.7 they are going to end up producing.

This is not a digression from the formal argument. It bears directly on what kind of specification a constraint-level description, in the sense of Chapter 2, can actually be expected to support. A specification that could only be realized by a constructor holding a complete mental image of the finished object's interior, planned out in advance down to every crease, would not really be sparse in the sense Chapter 3 claimed for it; the sparseness would be an illusion, with all the missing detail simply relocated into the constructor's head rather than genuinely left open. What this chapter argues is that the Supercube's construction did not require this, and that its actual method is more interesting, and more generally useful, than a mental substitute for the missing specification would have been.

7.2 Externalized visualization

The marked sheet itself carries the information a constructor needs to proceed, at any given moment, without requiring that constructor to hold a complete picture of the finished object in mind. The three squares, their edge annotations, and their shared vertex label remain physically present and physically inspectable throughout the construction. A constructor does not need to remember, in the sense of recalling from memory, which edge was meant to be the bottom edge; the mark is still there, on the sheet, wherever the sheet currently is, and can simply be looked at again.

This matters more than it might first appear to, because it changes what kind of cognitive demand the construction actually places on whoever is carrying it out. The demand is not: hold the complete target geometry in mind and manipulate the sheet to match it. The demand is narrower and more local: look at the currently visible marks, notice which declared relations are not yet satisfied, and make whatever move seems likely to bring one of them closer to being satisfied. The marked sheet functions as an external store of exactly the information that a complete mental image would otherwise have had to hold internally.

This point bears specifically, and should be stated carefully, on aphantasia, the reported absence or marked reduction of voluntary visual imagery in some individuals. It would be a significant overreach to suggest that this construction method demonstrates anything about the spatial reasoning capacities of people who report aphantasia, and no such claim is made here. What can be said, more modestly, is that a construction method built around externalized, physically inspectable marks and iterative local search does not appear to presuppose the kind of detailed, sustained voluntary visual imagery whose absence aphantasia describes. Whether or not a given constructor experiences vivid mental imagery of the finished cube while folding toward it, the marked sheet supplies, externally, the same information that imagery might otherwise have supplied internally. This is a claim about what the method requires, not a claim about what any particular person's imagery does or does not include, and it should not be read more broadly than that.

7.3 Local solvability

A further consequence of working from externalized marks rather than a complete internal plan is that the construction can proceed by addressing small, local questions one at a time, rather than requiring the global configuration to be solved all at once.

At any given moment in the Supercube's construction, a useful next move could be identified by asking a narrow question: can these two marked edges be brought closer together without disturbing a relation already secured? Can the corner that has already been formed be held steady while the material around it is rearranged? Can a bulge of surplus material, currently protruding where a marked face needs to sit flat, be pushed further inward? None of these questions requires an answer to the larger question of what the finished object will ultimately look like in its entirety. Each is answerable by inspecting the sheet's current state against the declared marks and judging, from that comparison alone, what the next useful move might be.

Global coherence, on this account, is not planned in advance and then executed. It accumulates, as a consequence of a sequence of locally useful moves, each one chosen for its immediate contribution to satisfying some currently unsatisfied relation, none of them requiring a synoptic view of the whole. This is worth stating plainly because it is a genuinely different model of how a complex spatial construction can be carried out than the model implicit in a conventional crease pattern, which does presuppose that the complete relationship between flat sheet and folded form is knowable, and known, before folding begins.

7.4 Failure as information

A fold that does not work, an edge that will not reach its declared partner, a corner that keeps slipping out of alignment as soon as attention moves elsewhere, is not, on the account developed here, an event external to the search process, an interruption to be gotten past as quickly as possible. It is itself a piece of information about the current shape of what Section 6.6 called $\mathrm{Reach}(P, h_t)$: the set of configurations still reachable given the sheet's history up to that point.

A blocked fold indicates one of a small number of specific things: that the surrounding material has become congested, in the sense of Section 6.5, and needs to be redistributed before the fold can proceed; that two regions have been brought into an orientation genuinely incompatible with the fold being attempted, requiring one or the other to be repositioned first; that insufficient slack remains in the surrounding sheet to permit the motion required, requiring surplus to be drawn in from elsewhere; or that a relation locked earlier, in the sense of Section 5.5, is now standing in the way of further progress and needs to be relaxed, per Section 5.6, before construction can continue. In each case, the failure is diagnostic. It reveals something true about the sheet's current reachable set that was not evident before the failed attempt was made, and it narrows the space of useful next moves rather than merely wasting the move that failed.

7.5 Repair and workaround

The natural response to a blocked fold, on this account, is not to undo the entire construction back to some earlier, unblocked state, though that remains available as a last resort. It is to introduce a local workaround: an additional pleat that provides the slack a straightforward fold could not; a pocket that temporarily holds a corner in place while attention moves to a different part of the sheet; a rotation of an already-folded bundle of material to relieve pressure against an adjacent, protected face; or a deliberate, bounded distortion, tolerated for the remainder of the construction, that trades a small amount of cosmetic regularity for continued progress toward the relations that actually matter.

These workarounds are local in the specific sense that they enlarge the reachable configuration space in the immediate vicinity of the obstruction without requiring the constructor to revisit or reconsider relations that have already been secured elsewhere on the sheet. This is what makes the process tractable by hand: a constructor need not hold the entire history of the construction in active consideration at every step, only the currently active obstruction and whatever already-secured relations that obstruction's resolution must not disturb.

7.6 Traces and self-scaffolding

Each fold made during construction leaves a trace, in the form of a crease, an established edge, or a temporary pocket, and these traces are available to guide subsequent action in a way that goes beyond simply recording that the fold occurred. An edge that has already been folded provides a stable reference against which a later fold can be aligned. A pocket formed to hold one corner in place can, incidentally, provide a convenient boundary against which an adjacent region can be pressed flat. The sheet, over the course of construction, progressively builds its own scaffolding: later moves are made easier, or made possible at all, by structure the earlier moves happened to leave behind.

This is a form of what is sometimes called stigmergic coordination in other contexts, where independent agents, or in this case a single agent acting over time, coordinate not through a shared plan held in common but through modifications each leaves in a shared environment that the others, or the agent's own later self, can perceive and respond to. No single moment of the Supercube's construction contained the complete solution. The solution, such as it was, emerged from the accumulated interaction among the declared markings, which never changed, the sheet's physical resistance, which shaped what could be attempted at each step, the traces left by earlier folds, which scaffolded later ones, and the ongoing comparison between the sheet's current state and the relations still outstanding.

This observation extends naturally, though only briefly needs to be extended here, to the possibility of multiple constructors, or multiple robotic manipulators, working on the same sheet without a complete shared plan. If each agent can perceive the declared marks and the traces left by prior actions, whether their own or another agent's, coordination of a genuinely useful kind becomes possible without requiring the agents to have agreed in advance on a complete sequence of operations. This possibility is noted here as a natural extension of the single-constructor account just given, rather than developed as an independent theory in its own right; the single-constructor case already contains everything essential to the point this chapter has been making, and multiplying the number of hands involved does not change the underlying logic of local, trace-guided, constraint-preserving search.

7.7 What the embodied account adds to the formal one

Chapter 5 gave a formal vocabulary, INFOLD, LOCK, RELAX, REPAIR, for describing constraint-directed search as a sequence of operations subject to declared conditions. This chapter has given the same process an experiential description, addressed to what it is actually like, and what it actually requires, to carry that search out by hand.

The two descriptions are not competing accounts of the same phenomenon so much as two different vantage points on it. The formal vocabulary specifies what a valid sequence of operations must accomplish, without saying anything about how a constructor is to discover such a sequence. This chapter has argued that the discovery process does not require, and in the Supercube's case did not involve, a complete internal image of the finished object; it required only a persistent external record of what mattered, a willingness to proceed by local, diagnostic, reversible steps, and a sheet whose own accumulated creases progressively scaffolded the steps that followed.

This has a direct bearing on the practical usefulness of constraint-directed infolding as a method, beyond its interest as a piece of formal theory. A specification that can be realized this way, sparse, externally inspectable, and searchable through local moves alone, is a specification that does not require its constructor to be an expert planner capable of anticipating a complete folding sequence in advance. It requires only the ability to compare a sheet's current state against a small number of declared marks and to recognize, moment to moment, whether the gap between them is narrowing. Chapter 8 turns to a different kind of surface facing a structurally similar problem, one where no constructor, human or otherwise, is available to carry out this kind of moment-to-moment comparison at all.

Part V

Biological Analogy

Chapter 8

Cortical Infolding Under Surface-Volume Constraint

8.1 A structural resemblance, stated cautiously

This chapter develops an analogy between the paper construction described in the preceding chapters and the folded structure of the mammalian cerebral cortex. It is important to state at the outset, before any of the resemblance is drawn out, exactly what kind of claim is and is not being made. The claim is structural: both systems involve a surface whose area substantially exceeds what could be smoothly accommodated within the volume that contains it, and both resolve this excess through folding rather than through simple expansion of the containing boundary. The claim is not mechanistic: this chapter does not propose, and does not believe, that the specific process by which a sheet of paper is folded by hand bears any direct resemblance to the specific biological processes by which a developing cortex actually folds. Those processes are addressed on their own terms in Section 8.3, and the differences between them and the paper construction are addressed directly and at comparable length in Section 8.4. The purpose of drawing the analogy at all is narrower than a claim of mechanism would be: it is to illustrate, using a case that can be inspected directly and folded by hand, a general geometric fact about area-rich surfaces in volume-limited environments that plausibly bears on the cortex without being confused with an account of how the cortex actually achieves it.

8.2 The shared problem

Stated abstractly, the shared problem is this: a surface possesses more area than the volume in which it is to be contained could accommodate if that surface remained smooth and unfolded. Something has to give. Either the containing volume expands to match the surface, or the surface's exposed extent is reduced by folding it into a smaller effective envelope, with the folded surface's full area preserved but no longer laid out flat.

The Supercube's sheet faced this problem directly, and by design: the three marked squares required a specific exterior configuration, and the sheet's substantial surplus area, defined formally in Section 6.4, had nowhere to go except into the interior the marked squares' configuration made available. The cerebral cortex, over the course of development, appears to face a structurally comparable problem: the cortical sheet's surface area increases considerably during a period in which the skull's interior volume is comparatively constrained, and a smooth, unfolded cortex of the area actually present in a mature human brain would not fit within a skull of ordinary size. Folding, in the form of the familiar pattern of gyri and sulci, the ridges and grooves visible on the brain's surface, allows substantially more cortical surface to exist within a given cranial volume than an unfolded sheet of the same area could.

This much is a shared geometric circumstance, and it is worth being clear that stating it this way commits to nothing about mechanism. It states only that both systems face an area-in-excess-of-volume problem and that both resolve it, whatever the process, by folding rather than by simple expansion.

8.3 An expansive metaphor, and the established mechanistic accounts

One evocative way to describe a surface's response to a spatial constraint imposed by its own growth, and by the growth of neighboring structures around it, borrows the phenomenon sometimes called crown shyness in forestry: the observation that the crowns of certain tree species, growing close to one another, tend to stop just short of touching, leaving a visible channel of open sky between adjacent canopies. This image is invoked here purely as a metaphor for constrained, mutually negotiated expansion into a bounded and already partially occupied space, and it should be understood as nothing more than that. Crown shyness is not itself an accepted account of cortical folding, and no claim is made that the mechanisms producing it in forest canopies bear any relation to the mechanisms producing folds in cortical tissue. The metaphor's only role is to evoke, vividly, the general idea of an expanding surface whose form is shaped by the presence of neighboring constraints rather than by its own growth in isolation, a general idea the cortex's situation shares even though the specific dynamics involved are entirely different.

The actual, established explanations for cortical gyrification are mechanistic and biological, and several distinct candidate mechanisms currently coexist in the literature without a single one having displaced the others as a complete account. Differential tangential expansion proposes that the cortical gray matter expands more rapidly, or along different constraints, than the white matter beneath it, producing a mechanical instability comparable to the buckling of a stiff, expanding layer bonded to a more slowly expanding substrate. Radial constraint accounts emphasize the geometric confinement imposed by the shape and growth rate of underlying structures. Axonal tension hypotheses propose that mechanical tension along developing white-matter connections pulls strongly interconnected cortical regions toward one another, contributing to the folding pattern. Additional accounts point to regional variation in cortical thickness, to the developmental sequencing and heterogeneity of different cortical areas, to the growth of underlying white matter volume, and to the geometry of the ventricular system as further contributing factors. These accounts are not mutually exclusive, and the current state of the relevant developmental biology treats gyrification as very likely multicausal, with different mechanisms contributing at different developmental stages and in different regions. Adjudicating among them is not the purpose of this chapter, and no attempt is made here to do so; they are listed to make clear that a substantial, ongoing, and specifically biological research program already exists to explain cortical folding, and that the paper analogy developed in this chapter is not offered as a contribution to that program or as a substitute for it.

8.4 What the analogy usefully captures

With the mechanistic accounts of Section 8.3 kept firmly separate, the structural analogy can be stated with some precision.

Both the folded sheet and the folded cortex preserve material continuity while changing extrinsic spatial relations. Just as Chapter 6 showed that the Supercube's three marked squares became exterior neighbors without any new material connection being formed between them, cortical regions that lie at some geodesic distance from one another along the intrinsic, unfolded cortical surface can end up in close three-dimensional proximity once the mature, folded cortex is examined, without any claim that this proximity reflects a direct developmental or connective relationship between those regions. Conversely, and just as significantly, two points that are close together on the folded exterior, sitting on either side of a sulcus, for instance, may be quite distant from one another along the cortical sheet's own intrinsic surface. Three-dimensional spatial proximity in a folded structure, whether paper or cortex, is not a reliable guide to intrinsic adjacency, and intrinsic adjacency is not a reliable guide to three-dimensional proximity.

This licenses a distinction, already familiar from the paper case, among several different notions of nearness that a cortical structure's geometry can support simultaneously and that do not, in general, coincide: material or intrinsic surface adjacency, three-dimensional geometric proximity after folding, connective or axonal proximity, developmental or lineage-based proximity, and functional or task-based association. The paper construction demonstrates cleanly, because it can be unfolded and directly inspected, that these several notions of nearness genuinely can diverge from one another once a surface has been folded; it offers this as a general geometric possibility available to any folded surface, cortex included, without asserting anything further about which of these notions of nearness the cortex's actual folding pattern happens to preserve or disrupt in any specific instance.

A further point of resemblance concerns history. Section 6.6 argued that a folded sheet's crease network functions as a partial, physically embedded record of its own folding history, and that this record constrains, without fully determining, what future configurations remain reachable. A mature, folded cortex likewise carries the residue of its own developmental history in its specific pattern of gyri and sulci, a pattern shaped by the developmental constraints, whichever combination of the mechanisms described in Section 8.3 turns out to have been operative, that were actually in force during its formation. And just as Section 6.7 showed that the same declared boundary specification could be reached by many different interior histories, it is a live possibility, consistent with current developmental biology though not established as fact by anything in this chapter, that similar mature folding patterns could arise from somewhat different combinations of the contributing developmental parameters, and that small differences arising early in development could be amplified into more substantial differences in the mature folding pattern. This is offered as a structural possibility the paper analogy makes vivid, not as an empirical claim about cortical development that this monograph is in a position to establish.

8.5 Where the analogy breaks

Several differences between the two systems are fundamental enough that the analogy should not be pressed past the structural point Section 8.4 draws from it, and each is worth stating plainly rather than left as an implicit qualification.

Paper is a passive, manufactured material, externally manipulated by an agent, the constructor, who exists outside the sheet and who folds it according to a plan, however sparse. Cortical tissue is living matter that grows, differentiates, remodels itself, develops its own vasculature, and interacts mechanically with the tissues surrounding it, all without an external agent folding it according to any plan at all; whatever organizing structure the process has is generated from within the developing tissue and its immediate environment, not imposed from outside by a hand holding a target specification.

The Supercube began with an explicit symbolic specification: three marked squares, each carrying declared edge roles and a declared shared vertex, fixed in advance of any folding and legible throughout the construction as an external reference the constructor could return to at any point. Cortical development does not appear to require, and there is no evidence that it involves, any comparable symbolic representation of the mature cortex's eventual folded form, held somewhere in advance of the folding process and consulted during it. Whatever determines the mature folding pattern is encoded, if the word encoded is even appropriate, in genetic, molecular, and mechanical factors operating locally and dynamically during development, not in anything resembling the three discrete, externally legible marks that organized the paper construction.

Uncontrolled crumpling, of the kind demonstrated in the alternative, more irregular states the Supercube's sheet was carried into in Section 6.5's discussion, can produce sharp ridges, localized regions of high curvature, and structural singularities that a passive sheet of paper tolerates without consequence but that living tissue plausibly could not, given constraints on cell viability, blood supply, and mechanical stress that a folded sheet of paper simply does not face. Cortical folding, whatever its precise mechanism, evidently proceeds under biological constraints on curvature, thickness, and structural integrity that have no counterpart in the paper case, and there is no reason to expect the geometric strategies available to an unconstrained sheet of paper to all be available to living cortical tissue.

8.6 A restrained conclusion

Given the qualifications of Section 8.5, the appropriate claim this chapter is entitled to make is considerably narrower than a reader might have expected on first encountering the analogy, and it is worth stating in a form precise enough to prevent the analogy from being carried further than this chapter intends.

> *The Crinkle-Cut Supercube is not a model of the mechanism of cortical gyrification. It is a model of the more general geometric fact that infolding allows an area-rich surface to inhabit a volume-limited environment while generating spatial relations absent from its unfolded presentation.*

Everything specific to how the cortex actually achieves this, which of the mechanisms surveyed in Section 8.3 predominates, in what combination, at what developmental stage, in what region, is a question for developmental biology to answer, and nothing in this chapter contributes to answering it. What the paper construction contributes is a directly inspectable, foldable, unfoldable demonstration that the general strategy, preserving material continuity while displacing surplus surface away from a constrained exterior, is geometrically available at all, and that a folded structure's exterior configuration, once achieved, will not by itself disclose the specific history or mechanism that produced it. This last point is the one thread that connects this chapter most directly back to the argument of Chapter 6, and it is the thread Chapter 9 now picks up in more general form, asking what can be said, formally, about when a target exterior configuration of this kind is achievable at all.

Part VI

General Principle and Consequences

Chapter 9

Continuation, Reachability, and the Crinkle-Cut Principle

9.1 Admissibility at each stage

At any moment during the construction of the Supercube, the sheet occupied some particular partially folded state, and from that state, some further moves were available and others were not. This is true trivially, in the sense that any physical process unfolds through a sequence of intermediate states, but it is worth making the observation precise, because the distinction between moves that remain available and moves that do not is what the rest of this chapter is built on.

Call a move **locally admissible** if making it does not irreparably violate a protected constraint, in the sense of Section 5.4, and does not damage the material integrity of the sheet itself, through tearing or an impermissible crease. A locally admissible move need not bring the construction closer to completion. It only needs to avoid closing off possibilities that were previously open.

9.2 Local admissibility is not enough

It is tempting to suppose that a construction proceeds successfully whenever each individual move, taken on its own, is locally admissible: if no single step damages anything already secured, the construction as a whole ought to succeed. This is false, and it is worth being precise about why, since the failure mode involved is exactly the kind of failure the Supercube's own construction had to navigate around.

A move can be locally admissible, in the narrow sense of Section 9.1, while still closing off the global path to completion. Bringing two of the three marked faces into close alignment, for instance, does not itself violate any protected constraint and may look, examined in isolation, like straightforward progress. But if that alignment traps a mass of surplus material in a position from which the third marked face cannot subsequently be brought into its required adjacency without either disturbing the first two or exceeding the sheet's capacity to bend without tearing, then the move was locally admissible and globally harmful at once. Nothing about examining the move on its own terms, at the moment it is made, reveals this consequence; the consequence only becomes visible once later moves are attempted and found to be blocked.

This distinction between local admissibility and **global reachability**, the continued existence of at least one path from the current state to a configuration satisfying every declared constraint, is the central technical idea of this chapter. A construction can proceed for a considerable time making only locally admissible moves and still arrive at a state from which no completion is reachable at all. Recognizing this possibility, and organizing the construction to guard against it, is a large part of what made the Supercube's own construction, as reconstructed in Chapter 1, proceed by establishing the vertex-gluing condition early and treating it as a stable anchor, in the sense of Section 5.5's LOCK operation, rather than by pursuing whichever local improvement happened to present itself first.

9.3 A hierarchy of protected distinctions

Because not every locally admissible move preserves global reachability, and because a construction carried out by hand cannot exhaustively verify global reachability before every move, some principle is needed for deciding which relations to protect most strongly when a choice has to be made between preserving one relation and preserving another. The Supercube's construction, examined in retrospect, appears to have followed an implicit ranking of exactly this kind, and it is useful to state that ranking explicitly:

$$\begin{gathered}\text{material continuity} > \text{selected-face identity} > \text{required incidence} \\ > \text{edge orientation} > \text{planarity} > \text{cosmetic regularity}\end{gathered}$$

Read from left to right, this hierarchy states that the sheet's basic material continuity, its status as a single connected surface without tears, is protected above everything else, since its loss would undermine the entire theoretical apparatus of Chapter 6, which depends on new exterior adjacency being achievable without new material connection. Below that, the identity of the three selected faces, which regions of the sheet count as the declared exterior at all, is protected more strongly than any specific relation among them. Below that again, the required incidence at the shared vertex and the required edge orientations are protected in turn, followed by the planarity of the individual faces, and finally by the cosmetic regularity of the finished object's overall appearance, which sits at the bottom of the hierarchy and was, in the Supercube's actual case, the distinction most freely sacrificed.

This hierarchy explains a fact that would otherwise be puzzling: why the finished Supercube, visibly rough and imperfect, nonetheless counts as a successful realization of its declared specification. Exact planarity, the fifth-ranked distinction, was sacrificed; cosmetic regularity, the sixth, was sacrificed considerably further. But material continuity, face identity, vertex incidence, and edge orientation, the four higher-ranked distinctions, were all preserved. Measured against the hierarchy rather than against an implicit standard of overall neatness, the construction achieved exactly what it was organized to achieve, and its visible roughness is the direct, expected consequence of having correctly prioritized higher-ranked distinctions over lower-ranked ones when a choice between them arose, rather than evidence of an unfinished or poorly executed construction.

9.4 Repair as renewed reachability

Section 5.6 introduced RELAX and REPAIR as operations permitting a locked relation to be temporarily loosened and subsequently restored. This chapter's distinction between local admissibility and global reachability gives those operations a sharper justification than mere convenience.

A repair succeeds, in the sense relevant here, not merely when it restores the relation's prior appearance, but when it reopens a path toward eventual completion that had been at risk of closing, while not sacrificing any distinction ranked above the one being repaired in the hierarchy of Section 9.3. A repair that restored a relation's cosmetic appearance while permanently foreclosing the construction's ability to satisfy a higher-ranked distinction elsewhere would not be a successful repair in this sense, whatever it looked like at the moment it was made. Repair, understood this way, is not restoration for its own sake. It is a targeted reopening of global reachability, undertaken at the specific point where reachability had become locally threatened, and evaluated by whether the threat is actually resolved rather than by whether the repaired region looks as it did before.

9.5 The finished object as a certificate

Once a construction reaches a state satisfying every constraint in its declared boundary grammar, that finished object stands as more than a completed artifact. It is a physical certificate that the declared specification was, in fact, satisfiable, given the sheet, the material, and the permitted operations actually available.

This is worth stating with some formality, because the distinction between demonstrating that a specification is satisfiable and merely conjecturing that it might be is exactly the distinction Chapter 1 drew, in its own terms, between a constructive existence claim and a general theorem. The Supercube's finished form does not certify that every non-contiguous three-face assignment is realizable; it certifies only that this one was, by the direct and irrefutable method of having actually been realized. The hidden folds inside the object, invisible from the exterior in exactly the sense Chapter 6 described, are the material trace of the successful search: a physical record, though not, per Section 6.8, a legible one, that some path through the space of admissible deformations was actually found and actually taken.

9.6 A formal statement: the Crinkle-Cut Principle

The preceding sections license a formal statement of what the Supercube's construction actually demonstrates, phrased with the deliberate restraint the word "some" is doing real work to provide:

> ***The Crinkle-Cut Principle.*** *For some connected sheets containing spatially separated, locally framed regions, a target exterior adjacency among those regions can be realized without cutting the sheet, when the complementary region is permitted sufficient infolding, overlap, curvature, and hidden interior occupation.*

The word "some" in this statement is not a placeholder for a stronger claim the monograph is unable to fully justify. It is the precise scope the constructive evidence actually supports. One sheet, with one specific assignment of three separated, framed regions, was successfully realized. This licenses the existence claim above, for that class of sheets and assignments broadly similar to the one demonstrated, and it licenses nothing stronger without further argument or further construction.

9.7 A stronger conjecture

A more ambitious claim can nonetheless be stated, clearly marked as a conjecture rather than as a result this monograph has established:

> ***Conjecture.*** *Broad families of finite boundary specifications are realizable from sufficiently large connected sheets, subject to constraints imposed by orientation,*

thickness, material continuity, edge length, self-contact, interior capacity, and tolerance.

This conjecture generalizes the Crinkle-Cut Principle from the demonstrated case of three separated square regions to a much wider class of possible boundary specifications, arbitrary numbers of regions, arbitrary shapes, arbitrary target adjacency graphs, while making explicit that realizability cannot be expected to hold unconditionally. The qualifying clause is not a formality. Section 9.8 takes up, in turn, several of the specific ways a boundary specification can fail to be realizable regardless of how much surplus surface and permitted infolding is made available, and these obstructions are exactly what the conjecture's qualifying clause is intended to acknowledge in advance.

9.8 Obstructions

Several distinct failure modes deserve to be named individually, since they arise for different reasons and would need to be ruled out separately in any attempt to establish the conjecture of Section 9.7 for a specific class of specifications.

A specification may demand **contradictory edge roles**: two edges each required, by the declared orientation conditions, to occupy roles that cannot simultaneously be satisfied by any consistent assignment of orientation to the intervening sheet. No amount of surplus material or permitted infolding resolves a contradiction of this kind, since the obstruction is combinatorial rather than geometric.

A specification may demand adjacency among regions separated by **insufficient connecting area**, requiring the intervening material to be compressed or displaced beyond what its actual area permits, regardless of how it is folded. This obstruction is geometric rather than combinatorial, and it interacts directly with the surplus ratio σ defined in Section 6.4: a specification with too low a surplus ratio, relative to what the declared adjacency and vertex conditions actually demand, may be unrealizable for reasons of raw material insufficiency alone.

A specification may be obstructed by **excessive material thickness**, where the sheet's actual physical thickness prevents the tight folding radii a given configuration requires, an obstruction with no counterpart in the idealized zero-thickness surface of Section 3.1 but a very real one for any physical sheet.

A specification may demand **impossible orientation requirements**, asking a region to be simultaneously reflected and unreflected relative to some other region, in a way no continuous deformation of a connected sheet can satisfy without tearing.

A specification may run into **inadequate interior volume**, where the target exterior's enclosed volume is simply too small to contain the surplus material the specification's surplus ratio requires to be relegated, an obstruction distinct from insufficient connecting area, since it concerns not whether the material can reach the interior but whether the interior can hold it once it arrives, closely related to the congestion discussed in Section 6.5.

A specification may produce a **self-locking trajectory**, in which every locally admissible sequence of moves available from some intermediate state leads to a configuration from which no further locally admissible move preserves global reachability, an obstruction that depends not on the specification alone but on the specific sequence of moves attempted, and that a differently-ordered construction, following Section 9.3's hierarchy more carefully, might have avoided even where a less careful one could not.

Finally, a specification may simply demand **geometric exactness incompatible with crumpling**, requiring a degree of precision in the final exterior that the permitted family of operations, which after all includes buckling and crumpling among its admissible deformations, cannot reliably deliver, even where a less exacting tolerance would have been readily achievable by the same means.

9.9 Ideal, mechanical, and practical realizability

It is worth closing this chapter by distinguishing three distinct senses in which a specification can be said to be realizable, since the obstructions of Section 9.8 do not all apply equally to each.

A specification is **ideally realizable** if some continuous deformation of an idealized, zero-thickness, infinitely flexible surface satisfies it, without regard to whether any physical material could actually achieve that deformation. A specification is **mechanically realizable** if some physically plausible deformation of a sheet with realistic thickness, stiffness, and plastic behavior satisfies it, accounting for the obstructions of Section 9.8 that arise specifically from material properties. A specification is **practically realizable** if a construction process, whether carried out by hand, by a computational search of the kind Chapter 10 briefly considers, or by some other method, can actually be found to locate a satisfying deformation within a reasonable amount of effort, distinct from the mere fact that such a deformation exists somewhere in the space of possibilities.

These three notions can, and in general do, come apart. A specification might be ideally realizable while being mechanically unrealizable for any sheet of nonzero thickness; it might be mechanically realizable while remaining practically elusive, its satisfying deformation existing but proving difficult for any actual search process, human or computational, to locate. The Supercube's own specification was demonstrated to be practically realizable, by the strongest possible evidence such a claim admits, an actual, physical, hand-executed construction that reached it. Nothing in this chapter should be read as extending that demonstrated practical realizability to any specification beyond the one actually constructed, and the Crinkle-Cut Principle of Section 9.6 is worded precisely to reflect this limit.

Chapter 10 closes the monograph's substantive argument by briefly indicating how the questions this chapter has posed, but deliberately not answered in general form, might be pursued further: through systematic variation of the specification, through computational search over the space of admissible deformations, and through comparison of independently constructed realizations of the same declared boundary grammar.

Chapter 10

Further Work

10.1 What remains open

The preceding chapters have established a constructive existence claim, a formal vocabulary adequate to state it precisely, a theoretical account of why a satisfied boundary specification cannot disclose the interior history that satisfied it, and a hedged principle, together with an explicit list of the ways that principle's stronger conjectured form could fail to hold. What they have not done, and do not attempt to do, is settle any of the questions that principle and conjecture leave open. This short chapter indicates, briefly and without developing any of them into a research program in their own right, the directions in which that settling might proceed.

10.2 Systematic variation

The most direct way to test how far the Crinkle-Cut Principle of Section 9.6 extends beyond the single demonstrated case is to vary the specification systematically and observe where realizability holds and where it fails. This means constructing further sheets with different numbers of selected regions, different relative placements and separations, different assigned edge orientations, and different surplus ratios, and attempting, by the same hand-folding methods described in Chapter 7, to realize each one. Success and failure should both be recorded, including the specific obstruction encountered in each failed attempt, drawn from the list in Section 9.8, since a pattern of which obstructions recur under which conditions would itself be informative about where the boundary of practical realizability actually lies.

A closely related line of variation concerns material rather than specification: repeating a fixed specification across sheets of differing thickness, stiffness, and plastic behavior, to determine how much of the boundary between success and failure is a property of the abstract specification and how much is a property of the particular material used to realize it. Section 9.9's distinction between ideal, mechanical, and practical realizability makes exactly this kind of variation necessary to disentangle, since a specification's mechanical realizability depends on material properties in a way its ideal realizability does not.

10.3 Independent realizations of the same specification

Chapter 6 argued, from a single sheet carried through two documented outcomes, that the same boundary specification can be satisfied by more than one interior history. A stronger version of this claim would ask several independent constructors, given the same specification and no knowledge of one another's methods, to realize it separately, and would then compare the resulting interiors, by the destructive unfolding or sectioning such a comparison would require, to determine how much genuine diversity of interior structure a single boundary class actually admits in practice, as opposed to in principle. This would give empirical content to the equivalence hierarchy introduced in Section 6.8, boundary, structural, crease, and procedural equivalence, by measuring how far apart independently produced realizations of the same specification actually tend to fall along that hierarchy.

10.4 Computational and robotic search

The INFOLD operator of Chapter 5 was framed throughout as a search instruction rather than a determinate procedure, and nothing in principle restricts that search to a human hand. A computational model could represent the selected regions as rigid or near-rigid panels connected by a deformable sheet, and search, by whatever optimization method proves suitable, for a configuration satisfying a declared boundary grammar while minimizing some combination of bending energy, self-collision, and interior congestion. Whether such a search reliably locates realizations that a human constructor would also find, or instead discovers structurally different solutions unavailable to hand-folding, would itself bear on the practical-realizability question raised in Section 9.9.

A robotic implementation, capable of perceiving the declared marks on a physical sheet and manipulating it directly, would test a further and more demanding question: whether the constraint-level notation developed in Chapter 4 carries enough information to guide a search process that has no access to the embodied, trace-guided intuitions described in Chapter 7, or whether some of what made the Supercube's construction tractable by hand depended on forms of tactile feedback and local judgment that a notation, however carefully specified, cannot fully substitute for.

10.5 A deliberate limit on this chapter's scope

None of the preceding directions is developed here beyond the brief indication given above. Detailed experimental templates, replication protocols, measurement instruments, and standardized recording forms belong to a separate, purpose-built replication supplement rather than to the body of this monograph, should such a supplement later be prepared. Including that material here would extend the monograph well past the scope this final part is meant to occupy, and would risk obscuring, under a mass of procedural detail, the more limited claim the monograph has actually been concerned to establish throughout: that one non-trivial constraint-directed infolding was successfully demonstrated, and that its formal and theoretical consequences, worked out across the preceding chapters, hold independently of whether any of the further work sketched in this chapter is ever carried out at all.

Conclusion: The Visible Solution

10.6 Returning to the object

It is worth returning, at the end of this monograph, to the object itself, rather than to the formal apparatus that has accumulated around it across the preceding chapters. Three squares were marked on a sheet considerably larger than the three of them combined. Each square carried a small amount of annotation: which edge was to be a bottom edge, which a side edge, which corner was to meet the corresponding corners of the other two squares at a single shared vertex. Nothing else about the sheet was specified in advance.

Nothing about the shape of the finished object was declared, in the sense that a cube net declares it, by laying out beforehand which regions must touch which. What was declared were fragments: three local frames and the relations required to hold among them. The rest of the sheet, unmarked and unaddressed by anything in the specification, was left to become whatever it needed to become in order for those fragments to be brought into their required configuration.

10.7 What the construction preserved and what it surrendered

The finished object honored the fragments. The three marked squares ended up mutually adjacent, correctly oriented, and incident to a shared corner, exactly as their annotations had specified. Everything else about the object, the exact route the surplus material took to reach the interior, the specific creases that route left behind, the object's overall neatness or lack of it, was never specified and was never, in any meaningful sense, decided in advance. It was discovered, by hand, through the kind of local, trace-guided search Chapter 7 described, and it could have been discovered differently, by a different sequence of moves, without the finished object thereby failing to satisfy the specification that governed it.

This is the sense in which the object's visible roughness is not a shortfall to be apologized for. Measured against the hierarchy of protected distinctions set out in Chapter 9, the construction preserved everything ranked above cosmetic regularity and sacrificed cosmetic regularity itself, which was, from the specification's own standpoint, exactly the correct trade to make. A cleaner object would not have been a more successful realization of this specification. It would have been evidence that a different, more demanding specification had been in force, one this monograph never claimed to be testing.

10.8 What generalizes and what does not

Sparse boundary constraints, this monograph has argued, can govern a physical transformation without determining its trajectory. This is not a paradox, once the distinction between a declared relation and a realized history, developed formally in Chapter 6 and experienced directly in Chapter 7, is kept in view. A specification that declares only a small number of relations leaves correspondingly many degrees of freedom open, and a physical sheet, permitted to fold, pleat, and crumple freely within those open degrees of freedom, will find some resolution of them, though not a predetermined one and not necessarily the same one twice.

The monograph has been careful, throughout, to keep this claim at the scale the evidence actually supports. One sheet, with one declared specification, was successfully realized, more than once, in more than one interior configuration, all of them satisfying the same declared exterior relations. This licenses the Crinkle-Cut Principle of Chapter 9 in its stated, deliberately narrow form, and it licenses the stronger conjecture stated alongside it only as a conjecture, not as a result this monograph has established. The cortical analogy of Chapter 8 was developed under the same discipline: a structural resemblance worth drawing out precisely because it is structural, and precisely because it was kept apart, explicitly and at comparable length, from any claim about mechanism that the paper construction cannot support and was never meant to.

10.9 The hidden disorder is not the opposite of the visible relation

The object's visible simplicity, three legible faces meeting correctly at a corner, and its hidden disorder, a densely folded and largely unrecoverable mass of surplus material, are not opposed to one another, and it would misread the entire argument of this monograph to treat them as though they were. The hidden disorder is not an unfortunate cost paid for the visible relation, tolerated because no better alternative was available. It is the specific mechanism by which the visible relation became achievable at all. A specification this sparse, governing regions this widely separated, could not have been realized by a sheet forbidden from crumpling; the freedom to crumple, and the resulting interior disorder that freedom produces, is not a defect the construction had to work around but the very capacity the construction depended on throughout.

This is the point at which the monograph's central formulation should be restated, now that the chapters supporting it have been laid out in full:

> *The boundary is not the whole object and does not contain its complete explanation. It is the visible residue of constraints successfully preserved through a larger, partly hidden history of deformation.*

10.10 A closing note on what has been offered

This monograph has offered a description, a notation, a theoretical account of what a satisfied boundary specification can and cannot disclose about its own history, a restrained biological analogy, and a formally hedged principle, together with an explicit account of where that principle's stronger form might fail. It has not offered a complete theory of paper mechanics, a general algorithm for realizing arbitrary boundary specifications, or a mechanistic account of cortical development, and it has said so plainly at every point where the temptation to claim more than this arose.

What it has offered, at bottom, is an argument built from a single physical object: that a boundary can be specified more sparsely than a complete crease pattern requires, that the gap between a sparse specification and a finished physical form can be closed by permitting unconstrained material to absorb whatever the specification leaves open, and that the resulting object, whatever else can be said about it, stands as a record, though not a legible one, of exactly this having been done. The three marked squares are still there, on the finished object, wherever it ends up. Everything else about it was found, not planned, and the finding is the part of the construction this monograph has tried, across its length, to take seriously as a subject in its own right.

Appendix A

Infolding Symbol Set

This appendix collects, in one place, the graphic notation introduced in Section 4.9. It is intended as a working reference for marking a physical sheet or a diagram of one, not as a restatement of the reasoning behind each symbol, which appears in the body of Chapter 4.

A.1 Protected faces

A selected region E_i is drawn with a heavy outline, distinguishing it at a glance from unmarked, free material. Each protected face is given a short identifier, F_1, F_2, F_3, and so on, placed inside or adjacent to the outlined region.

A.2 Free material

Regions belonging to the infoldable remainder I are left unmarked, or given only a light, uniform shading if some visual indication of extent is useful. The absence of a heavy outline is itself the operative signal: it declares that the region's internal crease structure lies outside the specification, not merely that it has not yet been drawn in.

A.3 Edge-role ticks

Edge roles are marked near the relevant edge, close to the outline of a protected face, using short ticks rather than the mountain- or valley-fold lines of conventional notation, which already carry a different meaning.

- A single solid tick marks an edge assigned the bottom role, B.
- A double tick marks an edge assigned the side role, S.

- A triple tick marks an edge assigned the top role, T, where a construction requires one.
- A matching pair of small letters, placed near two edges belonging to different protected faces, marks an adjacency condition $A(E_i{:}e_a,\, E_j{:}e_b)$ between them: the two edges carrying the same letter are the two required to meet.
- A small $\times$ near an edge marks that edge as prohibited from exterior exposure, the role X.

A.4 Vertex labels

A small circled label, such as V_1, placed at a corner of a protected face, indicates that this corner participates in a vertex-gluing condition. The same label, repeated at a corner of each of the other protected faces required to meet it, indicates that all corners carrying that label must become incident to one shared exterior vertex in the realized object. An unrepeated label, appearing at only one corner, carries no such requirement and should not be used; every vertex label in a complete diagram should appear at two or more corners.

A.5 Exposure status

Where a region's exposure status is not simply assumed from its being marked as protected or left free, a short label may be attached directly to the region:

- OUT, for a region intended to be principally exterior.
- IN, for a region intended to be principally interior.
- FREE, for a region where either status is acceptable to the specification.
- SEAM, for a region permitted to appear only as a narrow boundary strip or overlap, rather than as a full exterior face.

A.6 Phase annotations

Where a construction is more easily communicated in stages, brief phase labels may be written separately from the geometric marks, typically in the margin or on an accompanying line of text rather than on the sheet itself: for example, *Phase 1: realize* V_1, followed by *Phase 2: align the bottom-edge cycle*, followed by *Phase 3: relegate the remaining surplus*. These annotations record the order in which constraints are intended to be secured. They do not record, and should not be read as recording, the exact hand motions by which any one phase is to be carried out.

A.7 Compatibility note

None of the symbols above are intended to replace the standard mountain-fold, valley-fold, reversal, sink, and rotation symbols of conventional origami diagramming. A single diagram may combine both vocabularies freely: conventional symbols specifying a particular local operation, and the symbols above specifying what that operation is being performed in order to achieve or preserve. Section 4.10 discusses this compatibility at greater length.

Appendix B

Formal Definitions

This appendix gathers, without re-deriving, the formal terms introduced across Chapters 3 through 6 and Chapter 9, for reference. Each entry cites the section in which the term is originally motivated and explained.

Substrate (P). The connected material sheet on which a construction is carried out, treated for formal purposes as a connected, bounded two-dimensional surface initially lying flat in the plane. (Section 3.1)

Selected region (E_i). One of a finite collection $E = \{E_1, \ldots, E_n\}$ of regions marked in advance on the substrate, intended to remain legible on the finished object's exterior. (Section 3.2)

Infoldable complement (I). The set-theoretic complement $I = P \setminus E$: the portion of the substrate not selected for exterior legibility, whose final geometry is not declared in advance. Not to be understood as disposable or unimportant material. (Section 3.2)

Local frame (F_i). A pair $F_i = (E_i, o_i)$, where o_i is orientation data assigning target roles to some or all of the boundary features of E_i, independent of the region's current position or its distance from other selected regions. (Section 3.3)

Edge role. A symbolic label attached to an edge of a local frame, drawn from a vocabulary including B (bottom), S (side), T (top), J_k (joining edge, indexed), V_k (vertex-incident edge, indexed), and X (exposure-prohibited). (Section 3.5)

Partial specification. A local frame or boundary grammar in which some edges or relations are deliberately left unlabeled, constituting a positive declaration that the corresponding feature's final configuration is not being constrained, rather than an omission. (Section 3.5, Section 3.6)

Realization (f). A deformation carrying the marked substrate P into a folded state, evaluated against whether it satisfies a declared constraint set C. (Section 4.1)

Boundary grammar. The set of constraint types, exposure, adjacency, vertex-gluing, orientation, relegation, and closure, taken together, by which a finished object's exterior can be specified without specifying the deformation that produces it. (Section 4.8)

Exposure condition. The requirement $f(E_i) \subset \partial\Omega$, that a selected region ends up on the finished object's apparent exterior boundary, within a stated tolerance. (Section 4.2)

Adjacency condition. The requirement $A(E_i{:}e_a,\ E_j{:}e_b)$, that a specified edge of one selected region meet a specified edge of another in the realized exterior. (Section 4.3)

Vertex-gluing condition. The requirement $G(F_1, F_2, F_3) = V$, that designated corners of several local frames become incident to one shared exterior vertex. (Section 4.4)

Orientation condition. The requirement that a realized deformation carry a local frame's declared orientation into agreement with its target orientation, up to tolerance. (Section 4.5)

Relegation condition. The requirement $f(I) \subset \Omega \cup H$, that the infoldable complement's image lie within the object's interior or its hidden boundary layers H, rather than protruding as unaccounted-for exterior surface. (Section 4.6)

Closure condition. Any of a family of requirements governing how complete, stable, or functionally container-like the finished object must be; deliberately left as a family rather than a single formula, since adequate closure depends on the construction's purpose. (Section 4.7)

INFOLD$(I \mid C)$**.** The operator denoting constrained search: deform the free region I through any permitted combination of admissible operations such that the constraint set C is satisfied, or is not thereby rendered unsatisfiable. Denotes a search family, not a determinate procedure. (Section 5.1)

Protected geometry. Geometry corresponding directly to a declared constraint, which a realization must not sacrifice in the course of resolving anything else. (Section 5.4)

Sacrificial geometry. Geometry whose precise final shape and crease history are not protected, and whose specificity, though not whose material, may be surrendered in service of satisfying protected constraints elsewhere. (Section 5.4)

LOCK(C_k)**.** The operation of treating a currently satisfied constraint as provisionally fixed, so that subsequent operations are chosen or avoided specifically to preserve it. (Section 5.5)

RELAX(C_k, ε)**.** The operation of permitting a previously locked constraint to drift within a stated tolerance ε, undertaken to clear an obstruction elsewhere in the construction. (Section 5.6)

REPAIR(C_k)**.** The operation of reasserting a relaxed constraint, ideally within its original tolerance, once the obstruction that necessitated relaxation has been cleared. (Section 5.6)

Sheet graph (G_0)**.** A graph whose nodes correspond to marked regions and intervening portions of the flat, unfolded substrate, and whose edges record direct connection across the original planar surface. (Section 6.1)

Boundary graph (G_∂). A graph whose nodes correspond to regions visible on the finished exterior, and whose edges record adjacency among them in the realized object. (Section 6.1)

Surplus ratio (σ). The quantity $\sigma = \big(\mathrm{Area}(P) - \mathrm{Area}(E_{\mathrm{visible}})\big)/\mathrm{Area}(E_{\mathrm{visible}})$, measuring how much of the substrate's area lies outside the declared exterior, relative to the declared exterior's own area. (Section 6.4)

Reachable set (Reach(P, h_t)). The set of configurations reachable from the substrate's current state, given a specific deformation history h_t up to time t; a property of the pair of sheet and history, not of the sheet's current shape alone. (Section 6.6)

Boundary equivalence. Two realizations satisfy the same declared exposure, adjacency, orientation, vertex, and closure conditions, regardless of anything else about them. The weakest and, for this monograph's purposes, most important notion of equivalence among realizations. (Section 6.8)

Structural equivalence. Two realizations are boundary-equivalent and additionally share the same pattern of interior layer contacts. (Section 6.8)

Crease equivalence. Two realizations are structurally equivalent and additionally share the same crease network up to a stated tolerance. (Section 6.8)

Procedural equivalence. Two realizations were produced by the same sequence of operations in the same order; the strongest notion of equivalence among realizations, and the one closest to the reproducibility conventional sequential diagrams aim at. (Section 6.8)

Local admissibility. A move that does not irreparably violate a protected constraint and does not damage the substrate's material integrity, evaluated on its own terms without regard to its consequences for later moves. (Section 9.1)

Global reachability. The continued existence, from a given intermediate state, of at least one path to a configuration satisfying every declared constraint. Distinct from, and not implied by, a sequence of individually locally admissible moves. (Section 9.2)

Hierarchy of protected distinctions. The ranking material continuity > selected-face identity > required incidence > edge orientation > planarity > cosmetic regularity, used to decide which relation to preserve when preserving one requires sacrificing another. (Section 9.3)

Ideal realizability. A specification is ideally realizable if some continuous deformation of an idealized, zero-thickness, infinitely flexible surface satisfies it. (Section 9.9)

Mechanical realizability. A specification is mechanically realizable if some physically plausible deformation of a sheet with realistic thickness, stiffness, and plastic behavior satisfies it. (Section 9.9)

Practical realizability. A specification is practically realizable if a construction process can actually be found, within reasonable effort, to locate a satisfying deformation, distinct from the mere existence of one. (Section 9.9)

Appendix C

Open Problems

The following questions are left unresolved by this monograph. Each is stated with a pointer to the chapter in which the relevant background concept was introduced, but none is developed here beyond the statement given.

Realizability. Beyond the single demonstrated case, what is the actual scope of the conjecture stated in Section 9.7? Is there a decision procedure, even in principle, for determining whether a given boundary specification is realizable from a given substrate, or is the question undecidable in general once crumpling is admitted as a permitted operation? (Section 9.6, Section 9.7)

Minimum surplus. For a given target boundary specification, what is the smallest surplus ratio σ, as defined in Section 6.4, from which the specification remains realizable? Does this minimum depend only on the specification's combinatorial structure, or also on material properties in the sense of Section 9.9's distinction between ideal and mechanical realizability?

Internal complexity. Given a realized object, can a meaningful measure of its hidden interior's complexity be defined, and if so, does exterior regularity reliably predict a high value of that measure, as the tradeoff discussed in Section 6.5 would suggest, or are there realizations that combine a regular exterior with a comparatively simple interior?

Diversity of interiors. How many structurally distinct interiors, in the sense of the equivalence hierarchy of Section 6.8, can realize the same boundary specification in practice? The single sheet examined in this monograph demonstrated at least two; the general shape of this space, for a fixed specification, remains unexamined. (Section 10.3 proposes an experimental approach.)

Material scaling. How do the obstructions catalogued in Section 9.8, particularly those arising from thickness and plastic behavior, scale with the size of the substrate and the size of the target object? Does a sufficiently large sheet overcome obstructions that a smaller sheet of the same material cannot?

Verification. Given a finished, unopened object, what can be determined about whether it satisfies a claimed boundary specification without destructively unfolding it, and how does the difficulty of this verification compare to the difficulty of the original search that produced the object? (Section 9.9, Section 10.4)

Computational search. Does a computational solver, searching the space of admissible deformations directly, tend to discover the same class of realizations a human constructor finds by hand, or does it discover structurally different solutions unavailable to embodied, trace-guided search of the kind described in Chapter 7? (Section 10.4)

Robotic construction. Does the constraint-level notation of Chapter 4 carry sufficient information to guide a robotic manipulator lacking the tactile feedback described in Chapter 7, or does successful hand construction depend on forms of local judgment the notation does not, and perhaps cannot, fully capture? (Section 10.4)

The cortical analogy's limits. Nothing in this monograph establishes which, if any, of the mechanisms surveyed in Section 8.3 the structural analogy of Chapter 8 actually bears on beyond the general geometric point stated in Section 8.6. Whether the notion of a boundary grammar, suitably adapted, could contribute any precise, testable content to developmental biology, as opposed to serving only as an illustrative analogy, is left entirely open.

Expressive completeness. Does the notation developed in Chapter 4, together with the operators of Chapter 5, suffice to state every boundary specification of practical interest, or are there classes of desired exterior relations that resist expression in this vocabulary and require its extension?

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